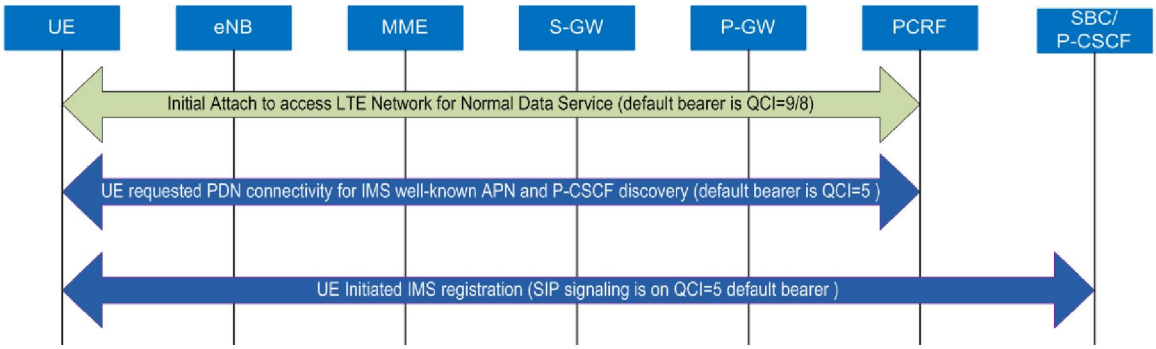
2 初始注册



1) VoLTE UE 在 LTE 网络内发起 Initial attach 流程，建立数据 APN 的 PDN connection，相应的 default EPS bearer 为 QCI9/8（由 HSS 上用户签约数据决定）；

   当核心网也支持 VoLTE 时，MME 会在 Attach 过程中告知 UE 网络侧支持 VoLTE。

2) VoLTE UE 发起到 IMS APN 的 PDN connection 建立流程，相应的缺省承载为 QCI5；在该过程中，获取 IMS 入口点 P-CSCF 地址信息。

3) VoLTE UE 发起到 IMS 域的注册，SIP 注册消息承载在 QCI5 缺省承载上。

2.1 Initial attach

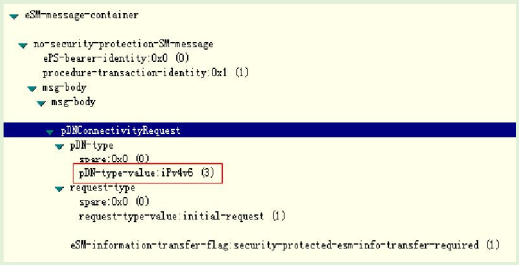
中移动终端会发起 combined attach 流程，在 Initial attach 过程中同时完成 CSFB 附着和普通数据业务附着。

数据 APN 采用 IPv4 地址，IMS APN 采用 IPv6 地址。

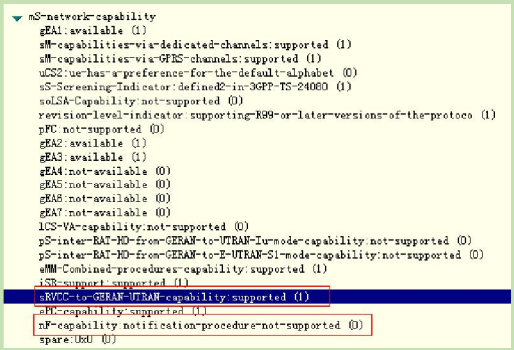
2.1.1 Attach Request（piggybacked PDN connectivity request）

Attach Request 消息关键 IE 截图：

关键 IE	IE 取值定义
attachRequest nAS-key-set-identifier tsc:native-security-context (0) nAS-key-set-identifier:nas-KSI2 (2) ePS-attach-type spare:0x0 (0) ePS-attach-type-value:combined-attach (2)	EPS 附着类型： 24.301 9.9.3.11 EPS attach type EPS attach type value (octet 1) Bits 3 2 1 0 0 1 EPS attach 0 1 0 combined EPS/IMSI attach 1 1 0 EPS emergency attach 1 1 1 reserved

<table>
<tr>
<td> 
</td>
<td>UE 请求的 IP 地址类型：

24.301 9.9.4.10 PDN type

PDN type value (octet 1)
Bits
3 2 1
0 0 1 IPv4
0 1 0 IPv6
0 1 1 IPv4v6
1 0 0 unused; shall be interpreted as "IPv6" if received by the network</td>
</tr>
<tr>
<td> 
</td>
<td>24.008 10.5.5.12 MS network capability

<MS network capability value part> ::=
<GEA1 bits>
<SM capabilities via dedicated channels: bit>
<SM capabilities via GPRS channels: bit>
<UCS2 support: bit>
<SS Screening Indicator: bit string(2)>
<SoLSA Capability : bit>
<Revision level indicator: bit>
<PFC feature mode: bit>
<Extended GEA bits>
<LCS VA capability: bit>
<PS inter-RAT HO from GERAN to UTRAN Iu mode capability: bit>
<PS inter-RAT HO from GERAN to E-UTRAN S1 mode capability: bit>
<EMM Combined procedures Capability: bit>
<ISR support: bit>
<SRVCC to GERAN/UTRAN capability: bit>
<EPC capability: bit>
<NF capability: bit>
<GERAN network sharing capability: bit>
<Spare bits>;</td>
</tr>
<tr>
<td>【解析】UE 上报 srvcc 能力和 Notification 能力给 MME。</td>
<td></td>
</tr>
<tr>
<td> 
</td>
<td>24.008 10.5.5.28 Voice domain preference and UE's usage setting

Voice domain preference and UE's usage setting value (octet 3, bit 1 to 3)

UE's usage setting (1 bit field)

Bit
3
0 Voice centric
1 Data centric

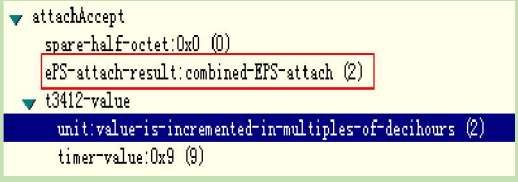
Voice domain preference for E-UTRAN (2 bit field)

Bit
2 1
0 0 CS Voice only
0 1 IMS PS Voice only
1 0 CS voice preferred, IMS PS Voice as secondary
1 1 IMS PS voice preferred, CS Voice as secondary

MS not supporting IMS voice shall indicate "CS Voice only".
MS only supporting IMS voice shall indicate "IMS PS Voice only".</td>
</tr>
<tr>
<td>【解析】终端上报 UE 的语音优先策略和终端属性</td>
<td></td>
</tr>
</table>

2.1.2 Attach Accept(piggybacked Activate default EPS bearer context request)

Attach Accept 关键 IE 截图：

<table>
<tr><th>关键 IE</th><th>IE 取值定义</th></tr>
<tr>
<td> 
</td>
<td>EPS 附着结果：

24.301 9.9.3.10 EPS attach result

EPS attach result value (octet 1)
Bits
3 2 1
0 0 1 EPS only
0 1 0 combined EPS/IMSI attach</td>
</tr>
<tr>
<td>【解析】这里表示 combined attach 成功</td>
<td></td>
</tr>
</table>

```
eSM-message-container
  no-security-protection-SM-message
    ePS-bearer-identity:0x5 (5)
    procedure-transaction-identity:0x1 (1)
    msg-body
      msg-body
        activateDefaultEPSBearerContextRequest
          ePS-QoS
            qCI:0x9 (9)
          access-point-name
            CHARACTER-STRING:cmnet
            CHARACTER-STRING:mnc007
            CHARACTER-STRING:mcc460
            CHARACTER-STRING:gprs
          pDN-address
            spare:0x0 (0)
            pDN-type-value:iPv4 (1)
            address-information:0A 4F 86 C2
```

【解析】该例中缺省承载为 QCI9，数据 APN 为 cmnet，为 UE 分配了 IPv4 地址。

24.301 9.9.4.9 PDN address

8	7	6	5	4	3	2	1	
PDN address IEI								octet 1
Length of PDN address contents								octet 2
0	0	0	0	0	PDN type value			octet 3
spare								
PDN address information								octet 4
								octet 15

Figure 9.9.4.9.1: PDN address information element

PDN type value (octet 3)

Bits

3	2	1	
0	0	1	IPv4
0	1	0	IPv6
0	1	1	IPv4v6

PDN address information (octet 4 to 15)

If PDN type value indicates IPv4, the PDN address information in octet 4 to octet 7 contains an IPv4 address. Bit 8 of octet 4 represents the most significant bit of the IPv4 address and bit 1 of octet 7 the least significant bit.

If PDN type value indicates IPv6, the PDN address information in octet 4 to octet 11 contains an IPv6 interface identifier. Bit 8 of octet 4 represents the most significant bit of the IPv6 interface identifier and bit 1 of octet 11 the least significant bit.

If PDN type value indicates IPv4v6, the PDN address information in octet 4 to octet 15 contains an IPv6 interface identifier and an IPv4 address. Bit 8 of octet 4 represents the most significant bit of the IPv6 interface identifier and bit 1 of octet 11 the least significant bit. Bit 8 of octet 12 represents the most significant bit of the IPv4 address and bit 1 of octet 15 the least significant bit.

If PDN type value indicates IPv4 or IPv4v6 and DHCPv4 is to be used to allocate the IPv4 address, the IPv4 address shall be coded as 0.0.0.0.

```
eps-network-feature-support
  spare:0x0 (0)
  cs-lcs:no-information-about-support-of-location-services-via-cs-domain-is-available (0)
  epc-lcs:location-services-via-epc-not-supported (0)
  emc-bs:emergency-bearer-services-in-s1-mode-not-supported (0)
  ims-vops:supported (1)
```

【解析】这里表示网络支持 IMS VoIP 业务。

24.301 9.9.3.12A EPS network feature support

8	7	6	5	4	3	2	1	
EPS network feature support IEI								octet 1
Length of EPS network feature support contents								octet 2
0	0	ESR	CS-LCS		EPC-LCS	EMC BS	IMS VoPS	octet 3
Spare		PS						

Figure 9.9.3.12A.1: EPS network feature support information element

IMS voice over PS session indicator (IMS VoPS) (octet 3, bit 1)

Bit

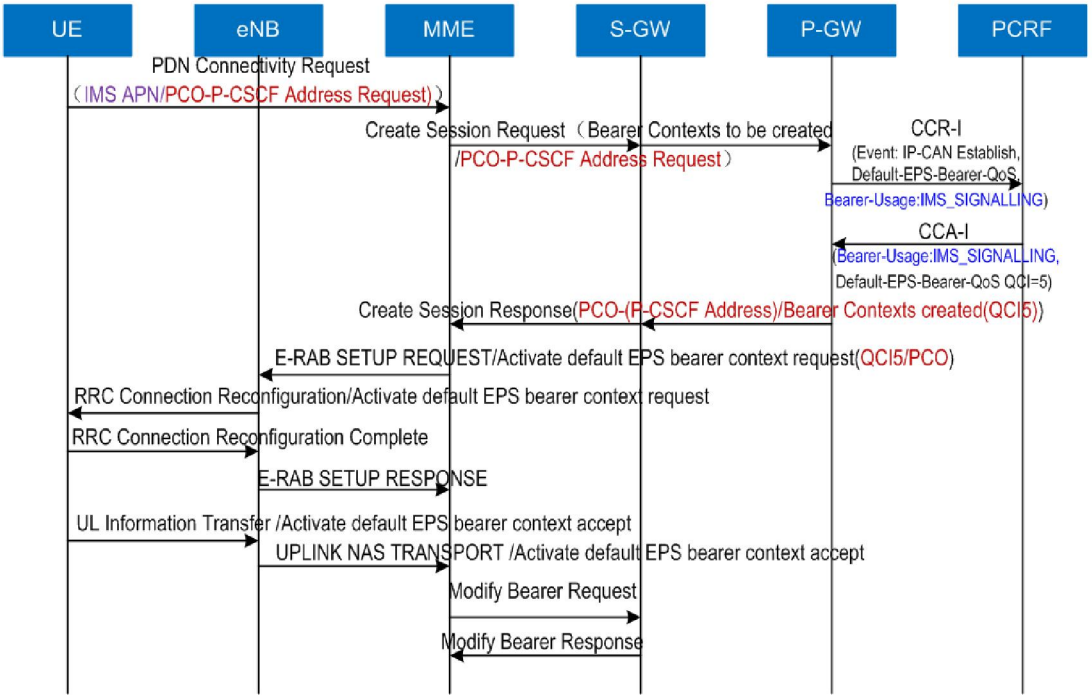
1	
0	IMS voice over PS session in S1 mode not supported
1	IMS voice over PS session in S1 mode supported

Emergency bearer services indicator (EMC BS) (octet 3, bit 2)

Bit

2	
0	emergency bearer services in S1 mode not supported
1	emergency bearer services in S1 mode supported

2.2 IMS PDN Connection 建立



2.2.1 PDN connectivity request

关键 IE	IE 取值定义
pDNConnectivityRequest pDN-type spare:0x0 (0) pDN-type-value:iPv4v6 (3) request-type spare:0x0 (0) request-type-value:initial-request (1) access-point-name CHARACTER-STRING:ims	24.301 9.9.4.10 PDN type PDN type value (octet 1) Bits 3 2 1 0 0 1 IPv4 0 1 0 IPv6 0 1 1 IPv4v6 1 0 0 unused; shall be interpreted as "IPv6" if received by the network 24.008 10.5.6.17 Request type Request type value (octet 1) Bits 3 2 1 0 0 1 initial request 0 1 0 Handover 0 1 1 Unused. If received, the network shall interpret this as "initial request". 1 0 0 emergency 24.008 10.5.6.1 Access point name
【解析】VoLTE UE 发起到 IMS APN PDN Connection 建立，APN IE 填写为 IMS well-known APN.	

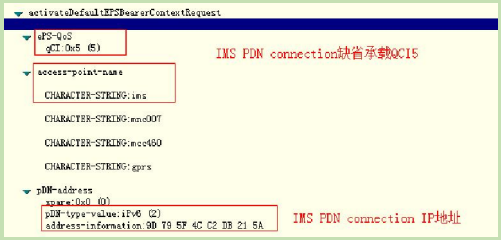
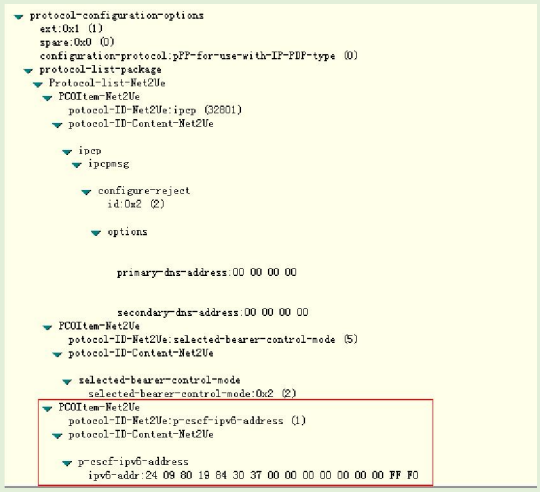
	8 7 6 5 4 3 2 1 Access point name IEI — octet 1 Length of access point name contents — octet 2 Access point name value — octet 3 … octet n* **Figure 10.5.152/3GPP TS 24.008: *Access point name* information element** The value part is defined in 3GPP TS 23.003 [10].
protocol-configuration-options ext:0x1 (1) spare:0x0 (0) configuration-protocol:PPP-for-use-with-IP-PDP-type (0) protocol-list-package Protocol-list-Ue2Net PCOItem-Ue2Net protocol-id-ue2net:ipcp (32801) protocol-id-content-ue2net ipcp ipcpmsg configure-request id:0x2 (2) options primary-dns-address:00 00 00 00 secondary-dns-address:00 00 00 00 PCOItem-Ue2Net protocol-id-ue2net:dns-server-ipv6-address-request (3) protocol-id-content-ue2net dns-server-ipv6-address-request:NULL PCOItem-Ue2Net protocol-id-ue2net:ip-address-allocation-via-nas-signalling (10) protocol-id-content-ue2net ip-address-allocation-via-nas-signalling:NULL PCOItem-Ue2Net protocol-id-ue2net:im-cn-subsystem-signaling-flag (2) protocol-id-content-ue2net im-cn-subsystem-signaling-flag:NULL PCOItem-Ue2Net protocol-id-ue2net:p-cscf-ipv4-address-request (12) protocol-id-content-ue2net p-cscf-ipv4-address-request:NULL PCOItem-Ue2Net protocol-id-ue2net:p-cscf-ipv6-address-request (1) protocol-id-content-ue2net p-cscf-ipv6-address-request:NULL PCOItem-Ue2Net protocol-id-ue2net:dns-server-ipv4-address-request (13) protocol-id-content-ue2net dns-server-ipv4-address-request:NULL 请求P-CSCF地址 【解析】终端请求 P-CSCF 地址	24.008 10.5.6.3 Protocol configuration options MS to network direction: - 0001H (P-CSCF IPv6 Address Request); - 0002H (IM CN Subsystem Signaling Flag); - 0003H (DNS Server IPv6 Address Request); - 0004H (Not Supported); - 0005H (MS Support of Network Requested Bearer Control indicator); - 0006H (Reserved); - 0007H (DSMIPv6 Home Agent Address Request; - 0008H (DSMIPv6 Home Network Prefix Request); - 0009H (DSMIPv6 IPv4 Home Agent Address Request); - 000AH (IP address allocation via NAS signalling); - 000BH (IPv4 address allocation via DHCPv4); - 000CH (P-CSCF IPv4 Address Request); - 000DH (DNS Server IPv4 Address Request); - 000EH (MSISDN Request); - 000FH (IFOM-Support-Request); - 0010H (IPv4 Link MTU Request); - 0011H (MS support of Local address in TFT indicator); and - FF00H to FFFFH reserved for operator specific use.

2.2.2 E-RAB SETUP REQUEST

关键 IE	IE 取值定义
e-RABToBeSetupItemBearerSUReq e-RAB-ID:0x6 (6) e-RABlevelQoSParameters qCI:0x5 (5) allocationRetentionPriority priorityLevel:0x1 (1): highest pre-emptionCapability:shall-not-trigger-pre-emption (0) pre-emptionVulnerability:pre-emptable (1) transportLayerAddress:01100100010000011111011010100001(64 41 FB 61) gTP-TEID:17 EF 97 D7 nAS-PDU NAS-MESSAGE security-protected-and-ciphered-NAS-message Activate default EPS bearer context request protected-nas:7C F5 12 3E 05 62 02 C1 01 05 17 B5 69 E9 73 06 6D 6E 63 30 30 37 06 6D 63 63 34 36 30 04 67 70 72 【解析】NAS PDU 对 eNodeB 透明	36.413 9.1.3.1 E-RAB SETUP REQUEST

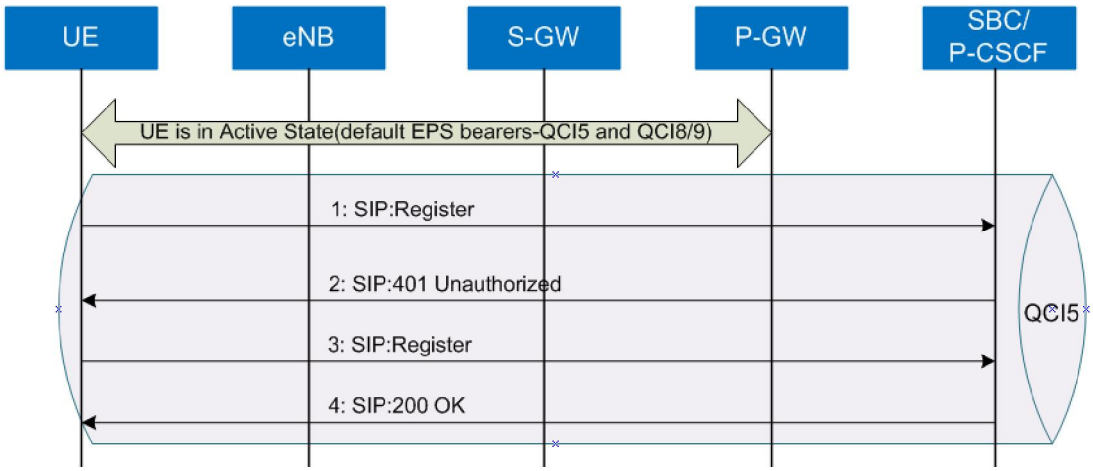
E-RAB to be Setup List		1	
>E-RAB To Be Setup Item IEs		1 .. <maxnoof E-RABs>	
>>E-RAB ID	M		9.2.1.2
>>E-RAB Level QoS Parameters	M		9.2.1.15
>>Transport Layer Address	M		9.2.2.1
>>GTP-TEID	M		9.2.2.2
>>NAS-PDU	M		9.2.3.5

2.2.3 Activate default EPS bearer context request

关键 IE	IE 取值定义
 【解析】激活 IMS PDN connection default EPS bearer（QCI5），并分配 IP 地址	PDN address
 【解析】MME 将从 P-GW 获取到的 P-CSCF 地址发送给 UE	24.008 10.5.6.3 Protocol configuration options Network to MS direction: - 0001H (P-CSCF IPv6 Address); - 0002H (IM CN Subsystem Signaling Flag); - 0003H (DNS Server IPv6 Address); - 0004H (Policy Control rejection code); - 0005H (Selected Bearer Control Mode; - 0006H (Reserved); - 0007H (DSMIPv6 Home Agent Address) ; - 0008H (DSMIPv6 Home Network Prefix); - 0009H (DSMIPv6 IPv4 Home Agent Address); - 000AH (Reserved); - 000BH (Reserved); - 000CH (P-CSCF IPv4 Address); - 000DH (DNS Server IPv4 Address); - 000EH (MSISDN); - 000FH (IFOM-Support); - 0010H (IPv4 Link MTU); - 0011H (Network support of Local address in TFT indicator); and - FF00H to FFFFH reserved for operator specific use.

2.3 IMS 域 SIP 注册

LTE 网络在 IMS 注册过程中，负责提供接入通道，负责消息转发，SIP 注册消息承载在 IMS PDN connection 的 default EPS bearer（QCI5）上，流程示意图如下：



GSMA IR.92 规定在 IMS 注册过程中，采用 IMS-AKA 算法完成 UE 和 IMS 网络之间的双向认证，认证成功后建立安全联盟 SA（Security Association），UE 和 P-CSCF 间 SIP 信令采用 IPsec 报文进去安全保护，完整性保护必选和加密可选：

2.2.2 Authentication

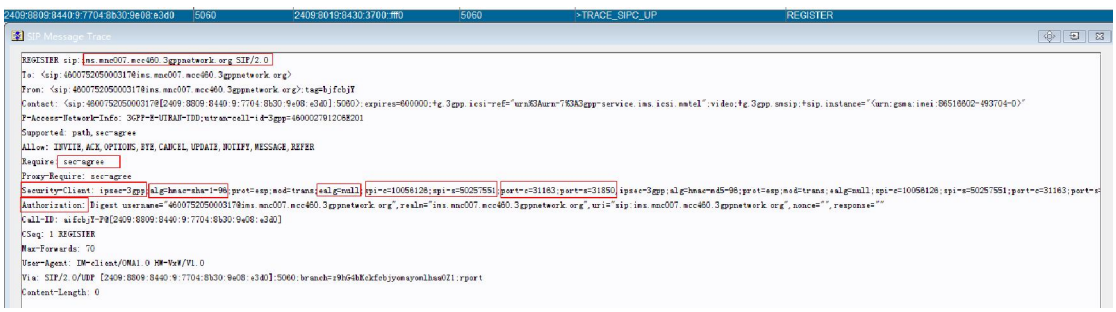
UE and IMS core network must follow the procedures defined in 3GPP TS 24.229 [15] and 3GPP TS 33.203 [45] for authentication with IMS Authentication and Key Agreement (IMS-AKA), Sec-Agree and IPSec. Support of integrity protection is mandatory for both UE and network. Support of confidentiality protection is optional in the network, considering that lower layer security is available.

如下为 P-CSCF（SE2900）信令截图：

TimeStamp	Source Address	Source Port	Destination Address	Destination Port	Message Interface Ty...	Message Type
2015-03-11 13:52:43.931	2409:8809:8440:c:428c:5d2e:b5a8:cc0c	5060	2409:8019:8430:3700::fff0	5060	>TRACE_SIPC_UP	REGISTER
2015-03-11 13:52:44.422	2409:8019:8430:3700::fff0	5060	2409:8809:8440:c:428c:5d2e:b5a8:cc0c	5060	<TRACE_SIPC_DOWN	401 UNAUTHORIZED
2015-03-11 13:52:45.022	2409:8809:8440:c:428c:5d2e:b5a8:cc0c	31162	2409:8019:8430:3700::fff0	9900	>TRACE_SIPC_UP	REGISTER
2015-03-11 13:52:45.421	2409:8019:8430:3700::fff0	9950	2409:8809:8440:c:428c:5d2e:b5a8:cc0c	31849	<TRACE_SIPC_DOWN	200 OK

下面以 P-CSCF 信令跟踪消息截图，简单说明关键消息 IE。

2.3.1 SIP Register

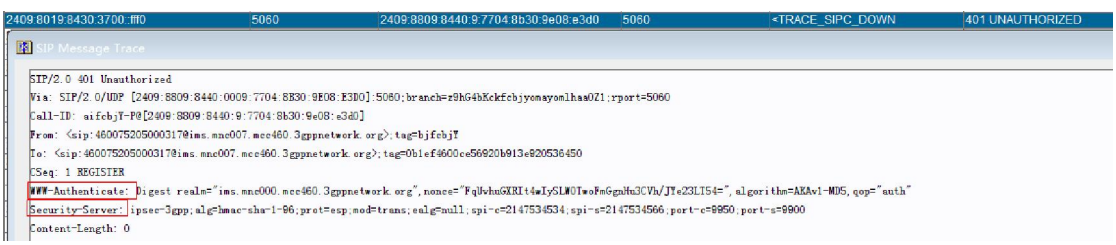


UE 读取 USIM 卡信息获取 IMSI，再从 IMSI 推导出 IMPI 和 IMPU，向 IMS 拜访网络入口 P-CSCF 发送 REGISTER 消息请求注册（在 SIP default port 5060 上发送）。

关键 IE	IE 取值定义
Request-URI REGISTER sip:ims.mnc007.mcc460.3gppnetwork.org SIP/2.0 【解析】用于指示注册位置服务的归属域名	RFC 3261 **10.2 Constructing the REGISTER Request** Request-URI: The Request-URI names the domain of the location service for which the registration is meant (for example, "sip:chicago.com"). The "userinfo" and "@" components of the SIP URI MUST NOT be present.
To: <sip:460075205000317@ims.mnc007.mcc460.3gppnetwork.org> From: <sip:460075205000317@ims.mnc007.mcc460.3gppnetwork.org>;tag=bbfcbjY 【解析】注册用户的公共身份标识 T-IMPU（Temporary IP Multimedia Public Identity）。	24.229 5.1.1.2.1 General a) a From header field set to the SIP URI that contains: 1) if the UE supports RFC 6140 [191] and performs the functions of an external attached network, the main URI of the UE; else 2) the public user identity to be registered; b) a To header field set to the SIP URI that contains: 1) if the UE supports RFC 6140 [191] and performs the functions of an external attached network, the main URI of the UE; else 2) the public user identity to be registered;
Contact: <sip:460075205000317@[2409:8809:8440:9:7704:8b30:9e08:e3d0]:5060>;expires=600000;+g.3gpp.icsi-ref="urn%3Aurn-7%3A3gpp-service.ims.icsi.mmtel";video;+g.3gpp.smsip;+sip.instance="<urn:gsma:imei:86516602-493704-0>" Contact 头域会携带注册用户的联系地址，注册时长。同时会携带一些标签表明自己支持业务的能力	RFC3261 **10.2 Constructing the REGISTER Request** Contact: REGISTER requests MAY contain a Contact header field with zero or more values containing address bindings. 24.229 5.1.1.2.1 General c) a Contact header field set to include SIP URI(s) containing the IP address or FQDN of the UE in the hostport parameter. If the UE: 1) supports GRUU (see table A.4, item A.4/53); 2) supports multiple registrations; 3) has an IMEI available; or 4) has an MEID available; e) a registration expiration interval value of 600 000 seconds as the value desired for the duration of the registration;
Supported: path,sec-agree Allow: INVITE,ACK,OPTIONS,BYE,CANCEL,UPDATE,NOTIFY,MESSAGE,REFER Require: sec-agree	RFC 3329 Security Mechanism Agreement for the Session Initiation Protocol (SIP) 24.229： 5.1.1.2.2 Initial registration using IMS AKA

Proxy-Require: sec-agree Security-Client: ipsec-3gpp;alg=hmac-sha-1-96;prot=esp;mod=trans;ealg=null;spi-c=10056126;spi-s=50257551;port-c=31163;port-s=31850 【解析】SIP 消息建立安全机制（IPSec-Security Association，指定 IPSec 完保(alg)和加密(ealg)算法）	d) a Security-Client header field set to specify the signalling plane security mechanism the UE supports, the IPsec layer algorithms the UE supports and the parameters needed for the security association setup. The UE shall support the setup of two pairs of security associations as defined in 3GPP TS 33.203 [19]. The syntax of the parameters needed for the security association setup is specified in annex H of 3GPP TS 33.203 [19]. The UE shall support the "ipsec-3gpp" security mechanism, as specified in RFC 3329 [48]. The UE shall support the IPsec layer algorithms for integrity and confidentiality protection as defined in 3GPP TS 33.203 [19], and shall announce support for them according to the procedures defined in RFC 3329 [48]. 33.203：7 Security association set-up procedure 33.203 The use of "Security Mechanism Agreement for SIP Sessions" [21] for security mode set-up security-client = "Security-Client" HCOLON sec-mechanism *(COMMA sec-mechanism) security-server = "Security-Server" HCOLON sec-mechanism *(COMMA sec-mechanism) security-verify = "Security-Verify" HCOLON sec-mechanism *(COMMA sec-mechanism) sec-mechanism = mechanism-name *(SEMI mech-parameters) mechanism-name = "ipsec-3gpp" / "tls" mech-parameters = (preference / algorithm / protocol / mode / encrypt-algorithm / spi-c / spi-s / port-c / port-s) preference = "q" EQUAL qvalue qvalue = ("0" ["." 0*3DIGIT]) / ("1" ["." 0*3("0")]) algorithm = "alg" EQUAL ("hmac-md5-96" / "hmac-sha-1-96") protocol = "prot" EQUAL ("ah" / "esp") mode = "mod" EQUAL ("trans" / "tun" / "UDP-enc-tun") encrypt-algorithm = "ealg" EQUAL ("des-ede3-cbc" / "aes-cbc" / "null") spi-c = "spi-c" EQUAL spivalue spi-s = "spi-s" EQUAL spivalue spivalue = 10DIGIT; 0 to 4294967295 port-c = "port-c" EQUAL port port-s = "port-s" EQUAL port port = 1*DIGIT
Authorization: Digest username="460075205000317@ims.mnc007.mcc460.3gppnetwork.org",realm="ims.mnc007.mcc460.3gppnetwork.org",uri="sip:ims.mnc007.mcc460.3gppnetwork.org",nonce="",response=""	24.229： 5.1.1.2.2 Initial registration using IMS AKA a) an Authorization header field, with: - the "username" header field parameter, set to the value of the private user identity; - the "realm" header field parameter, set to the domain name of the home network; - the "uri" header field parameter, set to the SIP URI of the domain name of the home network; - the "nonce" header field parameter, set to an empty value; and - the "response" header field parameter, set to an empty value; 33.203：6.1 Authentication and key agreement
Via:SIP/2.0/UDP [2409:8809:8440:9:7704:8b30:9e08:e3d0]:5060;branch=z9hG4bKckfcbjyomayomlhaaOZ1;rport	24.229 5.1.1.2.1 General d) a Via header field set to include the sent-by field containing the IP address or FQDN of the UE and the port number where the UE expects to receive the response to this request when UDP is used. For TCP, the response is received on the TCP connection on which the request was sent. For the UDP, the UE shall also include a "rport" header field parameter with no value in the Via header field. Unless the UE has been configured to not send keep-alives, and unless the UE is directly connected to an IP-CAN for which usage of NAT is not defined, it shall include a "keep" header field parameter with no value in the Via header field, in order to indicate support of sending keep-alives associated with the registration, as described in RFC 6223 [143]; NOTE 3: When sending the unprotected REGISTER request using UDP, the UE transmit the request from the same IP address and port on which it expects to receive the response to this request.

2.3.2 401 Unauthorized



关键 IE	IE 取值定义
WWW-Authenticate: Digest realm="ims.mnc000.mcc460.3gppnetwork.org",nonce="FqUvhuGXRIt4wIySLWOTwoFmGgnHu3CVh/JYe23LT54=",algorithm=AKAv1-MD5,qop="auth" 【解析】下发鉴权挑战	RFC 3261 20.44 WWW-Authenticate A WWW-Authenticate header field value contains an authentication challenge. See Section 22.2 for further details on its usage. 33.203 6.1.1 Authentication of an IM-subscriber RFC 3310 nonce A parameter, which is populated with the Base64 [4] encoding of the concatenation of the AKA authentication challenge RAND, the AKA AUTN token, and optionally some server specific data, as in Figure 1.
Security-Server: ipsec-3gpp;alg=hmac-sha-1-96;prot=esp;mod=trans;ealg=null;spi-c=2147534534;spi-s=2147534566;port-c=9950;port-s=9900 【解析】服务端建立 IPSec Association，分配保密端口作为传输层的源、目的端口	RFC 3329 Security Mechanism Agreement for the Session Initiation Protocol (SIP) 33.203：7 Security association set-up procedure

2.3.3 SIP Register

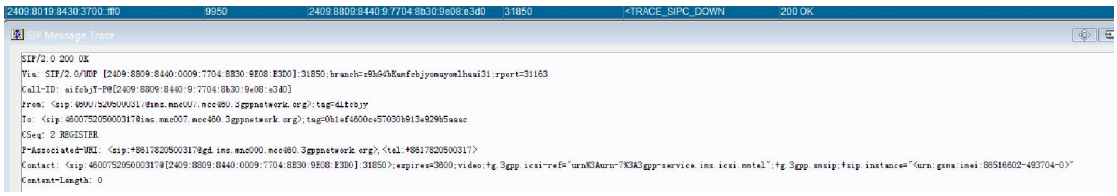


UE 收到 401 响应后，根据本地 USIM/ISIM 中保存的共享密钥对 AUTN 进行认证，认证通过则表明 401 消息来源于用户真实的归属网络，UE 对 IMS 网络鉴权通过。再基于共享密钥和 RAND 计算出 RES（Response），重新构造 REGISTER 消息，携带 RES，发给 P-CSCF。

关键 IE	IE 取值定义
Require: sec-agree Proxy-Require: sec-agree Security-Client: ipsec-3gpp;alg=hmac-sha-1-96;prot=esp;mod=trans;ealg=null;spi-	24.229 5.1.1.5.1 IMS AKA - general The UE shall also insert the Security-Client header field that is identical to the Security-Client header field that was included in the previous REGISTER request (i.e. the REGISTER request that was challenged with the received 401 (Unauthorized) response). The UE shall also insert the Security-Verify header field into the request, by mirroring in it the content of the Security-Server header field received in the 401 (Unauthorized) response.

c=10056126;spi-s=50257551;port-c=31163;port-s=31850,ipsec-3gpp;alg=hmac-md5-96;prot=esp;mod=trans;ealg=null;spi-c=10056126;spi-s=50257551;port-c=31163;port-s=31850 Security-Verify: ipsec-3gpp;alg=hmac-sha-1-96;prot=esp;mod=trans;ealg=null;spi-c=2147534534;spi-s=2147534566;port-c=9950;port-s=9900 【解析】建立 IPSec Association	
Authorization: Digest nonce="FqUvhuGXRIt4wlySLWOTwoFmGgnHu3CVh/JYe23LT54=",username="460075205000317@ims.mnc007.mcc460.3gppnetwork.org",uri="sip:ims.mnc007.mcc460.3gppnetwork.org",realm="ims.mnc000.mcc460.3gppnetwork.org",algorithm=AKAv1-MD5,response="e4c1070a1e2d1d11c82ef054687c71a2",qop=auth,cnonce="35402d0c35402cec",nc=00000001	24.229 5.1.1.5.1 IMS AKA - general 4) send another REGISTER request towards the protected server port indicated in the response using the temporary set of security associations to protect the message. The header fields are populated as defined for the initial REGISTER request that was challenged with the received 401 (Unauthorized) response, with the addition that the UE shall include an Authorization header field containing: - the "realm" header field parameter set to the value as received in the "realm" WWW-Authenticate header field parameter; - the "username" header field parameter, set to the value of the private user identity; - the "response" header field parameter that contains the RES parameter, as described in RFC 3310 [49]; - the "uri" header field parameter, set to the SIP URI of the domain name of the home network; - the "algorithm" header field parameter, set to the value received in the 401 (Unauthorized) response; and - the "nonce" header field parameter, set to the value received in the 401 (Unauthorized) response.

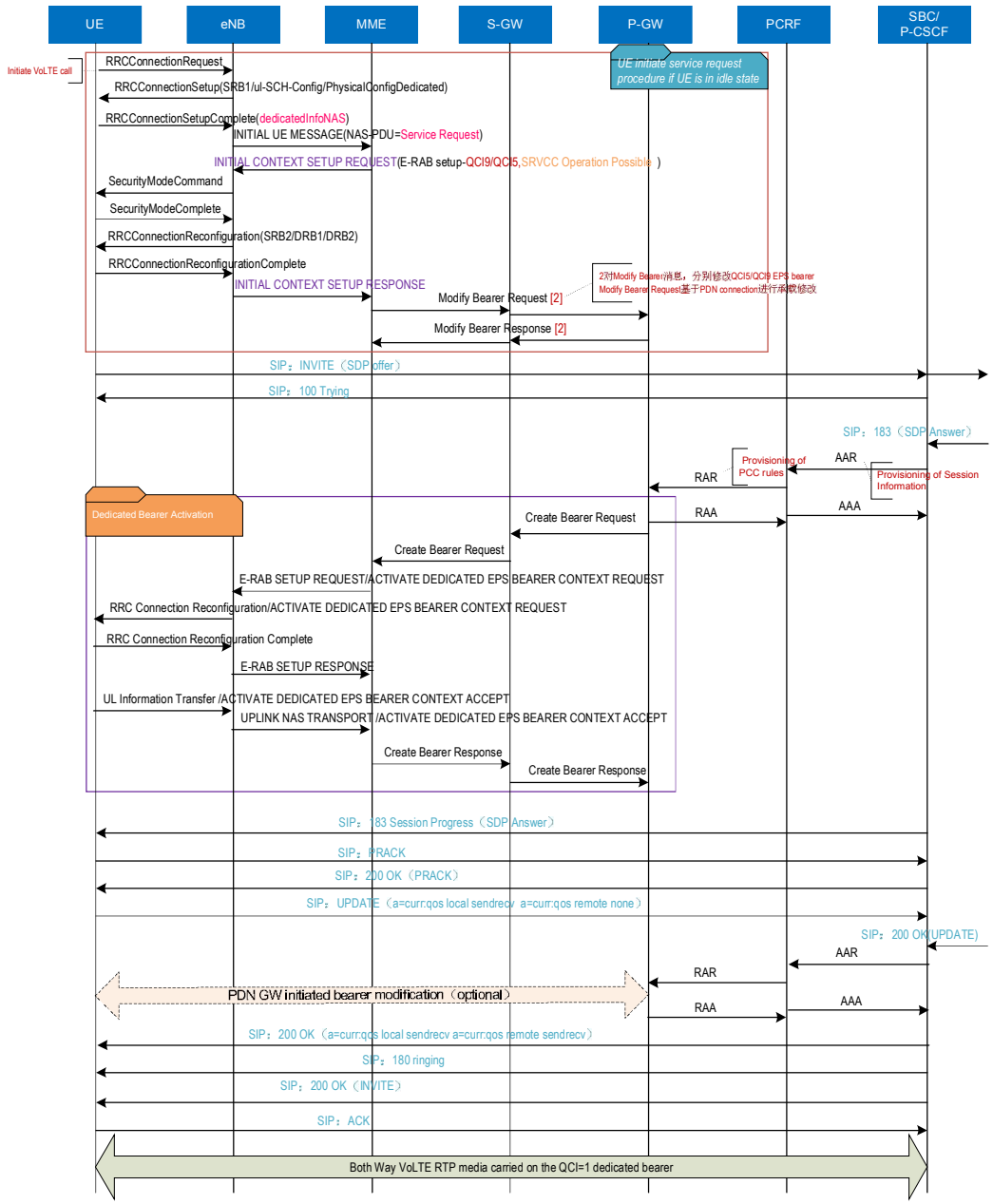
2.3.4 SIP 200 OK



P-CSCF 收到 UE 发送的 SIP Register 消息后，转发给 I-CSCF/S-CSCF， S-CSCF 收到鉴权响应，将期望收到的鉴权响应 XRES 和实际收到的鉴权响应 RES 进行比较。如果两者匹配，则 IMS 网络对 UE 鉴权通过。向 UE 回复 SIP 200 OK，表明初始注册成功。

3 VoLTE 呼叫 VoLTE 流程

3.1 主叫呼叫流程



eNodeB 虚用户信令跟踪：

RRC_CONN_REQ	Received From UE	mmec=6e; tmsi=c0 7e c9 04; RRCCause=mo-data;
RRC_CONN_SETUP	Send to UE	transmissionMode=tm3; SRS-Index=15; SRS-Bandwidth=bw2; CQI-A...
RRC_CONN_SETUP_CMP	Received From UE	
S1AP_INITIAL_UE_MSG	Send to MME	service request; enbs1apid=101686; RRCCause=mo-data; tai-PLMN=...
S1AP_INITIAL_CONTEXT_SETUP_REQ	Received From MME	eRABID=6,5; qci=5,9; release=rel9; ue-CAT=4; FGI=7f cf de bc; SRVC...
RRC_SECUR_MODE_CMD	Send to UE	
RRC_SECUR_MODE_CMP	Received From UE	
RRC_CONN_RECFG	Send to UE	cqi-Aperiodic=rm30;
RRC_CONN_RECFG_CMP	Received From UE	
S1AP_INITIAL_CONTEXT_SETUP_RSP	Send to MME	
RRC_UE_CAP_ENQUIRY	Send to UE	
RRC_CONN_RECFG	Send to UE	AddMearID=1,2,3,4,5,6,7; AddMearObjID=1,1,1,1,1,1,1; AddRptID=1,...
RRC_UE_CAP_INFO	Received From UE	
RRC_CONN_RECFG_CMP	Received From UE	
RRC_MEAS_RPRT	Received From UE	MSID=7; servRSRP=-98; servRSRQ=-5; neighber(PCI=24,RSRP=-108,...
RRC_MEAS_RPRT	Received From UE	MSID=5; servRSRP=-98; servRSRQ=-5;
RRC_CONN_RECFG	Send to UE	cqi-Aperiodic=rm30;
RRC_CONN_RECFG_CMP	Received From UE	
RRC_MEAS_RPRT	Received From UE	MSID=2; servRSRP=-98; servRSRQ=-6;
RRC_CONN_RECFG	Send to UE	
RRC_CONN_RECFG_CMP	Received From UE	
S1AP_ERAB_SETUP_REQ	Received From MME	eRABID=7; qci=1; IpAddress=100.65.251.210;
RRC_CONN_RECFG	Send to UE	
RRC_CONN_RECFG_CMP	Received From UE	
S1AP_ERAB_SETUP_RSP	Send to MME	
RRC_UL_INFO_TRANSF	Received From UE	
S1AP_UL_NAS_TRANS	Send to MME	
RRC_CONN_RECFG	Send to UE	AddMearID=7; AddMearObjID=1; AddRptID=7; freq=39250; pci=21;
RRC_CONN_RECFG_CMP	Received From UE	
RRC_CONN_RECFG	Send to UE	
RRC_CONN_RECFG_CMP	Received From UE	

SBC 侧信令跟踪：

2409:8809:8440:9:7704:8b30:9e08:e3d0	31163	2409:8019:8430:3700::fff0	9900	>TRACE_SIPC_UP	INVITE
2409:8019:8430:3700::fff0	9950	2409:8809:8440:9:7704:8b30:9e08:e3d0	31850	<TRACE_SIPC_DOWN	100 TRYING
10.132.170.2	16997	10.131.158.2	5060	<TRACE_SIPC_DOWN	INVITE
10.131.158.2	5060	10.132.170.2	16997	>TRACE_SIPC_UP	100 TRYING
10.131.158.2	5060	10.132.170.2	16996	>TRACE_SIPC_UP	INVITE
10.132.170.2	16996	10.131.158.2	5060	<TRACE_SIPC_DOWN	100 TRYING
2409:8019:8430:3700::fff0	9900	2409:8809:8440:3:cdf8:f326:ac61:d29e	31846	<TRACE_SIPC_DOWN	INVITE
2409:8809:8440:3:cdf8:f326:ac61:d29e	31846	2409:8019:8430:3700::fff0	9950	>TRACE_SIPC_UP	100 TRYING
2409:8809:8440:3:cdf8:f326:ac61:d29e	31846	2409:8019:8430:3700::fff0	9950	>TRACE_SIPC_UP	183 SESSION PROGRESS
10.132.170.2	16996	10.131.158.2	5060	<TRACE_SIPC_DOWN	183 SESSION PROGRESS
10.131.158.2	5060	10.132.170.2	16997	>TRACE_SIPC_UP	183 SESSION PROGRESS
10.127.78.61	3869	10.127.71.46	7001	TRACE_DIAM_GQ	AAR
10.127.71.46	7001	10.127.78.61	3869	TRACE_DIAM_GQ	AAA
2409:8019:8430:3700::fff0	9950	2409:8809:8440:9:7704:8b30:9e08:e3d0	31850	<TRACE_SIPC_DOWN	183 SESSION PROGRESS
2409:8809:8440:9:7704:8b30:9e08:e3d0	31163	2409:8019:8430:3700::fff0	9900	>TRACE_SIPC_UP	PRACK
10.132.170.2	16997	10.131.158.2	5060	<TRACE_SIPC_DOWN	PRACK
10.131.158.2	5060	10.132.170.2	16996	>TRACE_SIPC_UP	PRACK
2409:8019:8430:3700::fff0	9900	2409:8809:8440:3:cdf8:f326:ac61:d29e	31846	<TRACE_SIPC_DOWN	PRACK
2409:8809:8440:3:cdf8:f326:ac61:d29e	31157	2409:8019:8430:3700::fff0	9900	>TRACE_SIPC_UP	200 OK
10.132.170.2	16996	10.131.158.2	5060	<TRACE_SIPC_DOWN	200 OK
10.131.158.2	5060	10.132.170.2	16997	>TRACE_SIPC_UP	200 OK
2409:8019:8430:3700::fff0	9950	2409:8809:8440:9:7704:8b30:9e08:e3d0	31850	<TRACE_SIPC_DOWN	200 OK
2409:8809:8440:9:7704:8b30:9e08:e3d0	31163	2409:8019:8430:3700::fff0	9900	>TRACE_SIPC_UP	UPDATE
10.132.170.2	16997	10.131.158.2	5060	<TRACE_SIPC_DOWN	UPDATE
10.131.158.2	5060	10.132.170.2	16996	>TRACE_SIPC_UP	UPDATE
2409:8019:8430:3700::fff0	9900	2409:8809:8440:3:cdf8:f326:ac61:d29e	31846	<TRACE_SIPC_DOWN	UPDATE
2409:8809:8440:3:cdf8:f326:ac61:d29e	31846	2409:8019:8430:3700::fff0	9950	>TRACE_SIPC_UP	200 OK
10.132.170.2	16996	10.131.158.2	5060	<TRACE_SIPC_DOWN	200 OK
10.131.158.2	5060	10.132.170.2	16997	>TRACE_SIPC_UP	200 OK
10.127.78.60	3868	10.127.71.42	7000	TRACE_DIAM_GQ	AAR
10.127.71.42	7000	10.127.78.60	3868	TRACE_DIAM_GQ	AAA
2409:8019:8430:3700::fff0	9950	2409:8809:8440:9:7704:8b30:9e08:e3d0	31850	<TRACE_SIPC_DOWN	200 OK
2409:8809:8440:3:cdf8:f326:ac61:d29e	31846	2409:8019:8430:3700::fff0	9950	>TRACE_SIPC_UP	180 RINGING
10.132.170.2	16996	10.131.158.2	5060	<TRACE_SIPC_DOWN	180 RINGING
10.131.158.2	5060	10.132.170.2	16997	>TRACE_SIPC_UP	180 RINGING
2409:8019:8430:3700::fff0	9950	2409:8809:8440:9:7704:8b30:9e08:e3d0	31850	<TRACE_SIPC_DOWN	180 RINGING
2409:8809:8440:3:cdf8:f326:ac61:d29e	31846	2409:8019:8430:3700::fff0	9950	>TRACE_SIPC_UP	200 OK
10.132.170.2	16996	10.131.158.2	5060	<TRACE_SIPC_DOWN	200 OK
10.131.158.2	5060	10.132.170.2	16997	>TRACE_SIPC_UP	200 OK
2409:8019:8430:3700::fff0	9950	2409:8809:8440:9:7704:8b30:9e08:e3d0	31850	<TRACE_SIPC_DOWN	200 OK
2409:8809:8440:9:7704:8b30:9e08:e3d0	31163	2409:8019:8430:3700::fff0	9900	>TRACE_SIPC_UP	ACK
10.132.170.2	16997	10.131.158.2	5060	<TRACE_SIPC_DOWN	ACK
10.131.158.2	5060	10.132.170.2	16996	>TRACE_SIPC_UP	ACK
2409:8019:8430:3700::fff0	9900	2409:8809:8440:3:cdf8:f326:ac61:d29e	31846	<TRACE_SIPC_DOWN	ACK

相关详细消息介绍如下：

3.1.1 RRCConnectionRequest

Idle 态 UE 因有上层业务需求，发起 RRC 建立流程，发送 RRCConnectionRequest 消息。

关键 IE	IE 取值定义

▼ ue-Identity
▼ s-TMSI
mmec:01101110(6E)
m-TMSI:11000000011111101100100100000100(C0 7E C9 04)
【解析】携带 UE ID 信息
	36.331 RRCConnectionRequest
	RRCConnectionRequest-r8-IEs ::= SEQUENCE {
	ue-Identity InitialUE-Identity,
	establishmentCause EstablishmentCause,
	spare BIT STRING (SIZE (1))
	}
	InitialUE-Identity ::= CHOICE {
	s-TMSI S-TMSI,
	randomValue BIT STRING (SIZE (40))
	}
	EstablishmentCause ::= ENUMERATED {
	emergency, highPriorityAccess, mt-Access, mo-Signalling,
	mo-Data, delayTolerantAccess-v1020, spare2, spare1}
	24.301 5.3.1.1 Establishment of the NAS signalling connection
	【说明】规定 NAS 层什么场景下，提供 S-TMSI 信息给 AS 层
	24.301 D.1 Mapping of NAS procedure to RRC establishment cause (S1 mode only)
	【说明】规定 NAS 流程与 RRC establishmentcause 取值对应关系

establishmentCause:mo-Data (4) 【解析】携带 RRC 建立原因值	

3.1.2 RRCConnectionSetup

eNodeB 发送 RRCConnectionSetup 消息，建立 SRB1，配置 ul-SCH-Config，PhysicalConfigDedicated（pdsch-ConfigDedicated/ pucch-ConfigDedicated/ schedulingRequestConfig...）等信息。

关键 IE	IE 取值定义

```
▼ SRB-ToAddMod

   srb-Identity:0x1 (1)
 ▼ rlc-Config

  ▼ explicitValue


    ▼ am
     ▼ ul-AM-RLC
         t-PollRetransmit:ms45 (8)
         pollPDU:pInfinity (7)
         pollByte:kBinfinity (14)

         maxRetxThreshold:t32 (7)
      ▼ dl-AM-RLC
         t-Reordering:ms35 (7)

         t-StatusProhibit:ms0 (0)
 ▼ logicalChannelConfig

  ▼ explicitValue

   ▼ ul-SpecificParameters

       priority:0x1 (1)

       prioritisedBitRate:infinity (7)
       bucketSizeDuration:ms300 (3)
       logicalChannelGroup:0x0 (0)
```

36.331 RadioResourceConfigDedicated

```
SRB-ToAddMod ::=    SEQUENCE {
    srb-Identity                    INTEGER (1..2),
    rlc-Config                      CHOICE {
        explicitValue                   RLC-Config,
        defaultValue                    NULL
    }       OPTIONAL,                                                   -- Cond Setup
    logicalChannelConfig            CHOICE {
        explicitValue                   LogicalChannelConfig,
        defaultValue                    NULL
    }       OPTIONAL,                                                   -- Cond Setup
    ...
}
```

36.331 RLC-Config

36.331 LogicalchannelConfig

```
▼ ul-SCH-Config


   maxHARQ-Tx:n5 (4)
   periodicBSR-Timer:sf10 (1)
   retxBSR-Timer:sf320 (0)
   ttiBundling:FALSE
```

36.331 MAC-MainConfig

```
ul-SCH-Config                   SEQUENCE {
    maxHARQ-Tx                      ENUMERATED {
                                        n1, n2, n3, n4, n5, n6, n7, n8,
                                        n10, n12, n16, n20, n24, n28,
                                        spare2, spare1}     OPTIONAL,   -- Need ON
    periodicBSR-Timer               ENUMERATED {
                                        sf5, sf10, sf16, sf20, sf32, sf40, sf64, sf80,
                                        sf128, sf160, sf320, sf640, sf1280, sf2560,
                                        infinity, spare1}   OPTIONAL,   -- Need ON
    retxBSR-Timer                   ENUMERATED {
                                        sf320, sf640, sf1280, sf2560, sf5120,
                                        sf10240, spare2, spare1},
    ttiBundling                     BOOLEAN
}                                                               OPTIONAL,   -- Need ON
```

```
▼ pucch-ConfigDedicated

 ▼ ackNackRepetition


    release:NULL
   tdd-AckNackFeedbackMode:bundling (0)
```

36.331 PhysicalConfigDedicated

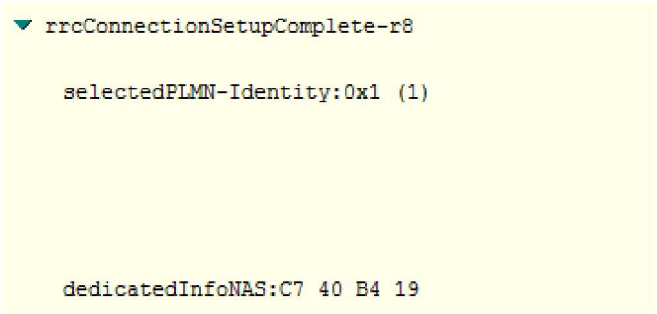
```
PhysicalConfigDedicated ::=     SEQUENCE {
    pdsch-ConfigDedicated           PDSCH-ConfigDedicated       OPTIONAL,       -- Need ON
    pucch-ConfigDedicated           PUCCH-ConfigDedicated       OPTIONAL,       -- Need ON
    pusch-ConfigDedicated           PUSCH-ConfigDedicated       OPTIONAL,       -- Need ON
    uplinkPowerControlDedicated     UplinkPowerControlDedicated OPTIONAL,       -- Need ON
    tpc-PDCCH-ConfigPUCCH           TPC-PDCCH-Config            OPTIONAL,       -- Need ON
    tpc-PDCCH-ConfigPUSCH           TPC-PDCCH-Config            OPTIONAL,       -- Need ON
    cqi-ReportConfig                CQI-ReportConfig            OPTIONAL,       -- Cond CQI-
r8
    soundingRS-UL-ConfigDedicated   SoundingRS-UL-ConfigDedicated OPTIONAL,     -- Need ON
    antennaInfo                     CHOICE {
        explicitValue                   AntennaInfoDedicated,
        defaultValue                    NULL
    }       OPTIONAL,                                                   -- Cond AI-r8
    schedulingRequestConfig         SchedulingRequestConfig     OPTIONAL,       -- Need ON
```

36.331 PUCCH-Config



3.1.3 RRCConnectionSetupComplete

终端回复 RRCConnectionSetupComplete 消息，携带 selectedPLMN-Identity 和 dedicatedInfoNAS（对应 Initial NAS Message，这里为 Service Request 消息）。



3.1.4 INITIAL UE MESSAGE

eNodeB 将在 RRCConnectionSetupComplete 消息中收到的 NAS 消息（dedicatedInfoNAS IE），通过 Initial UE Message 消息发送给 MME，同时携带 UE 接入当前小区的 TAI，ECGI，RRC Establishment Cause，S-TMSI 等信息。

3.1.5 INITIAL CONTEXT SETUP REQUEST

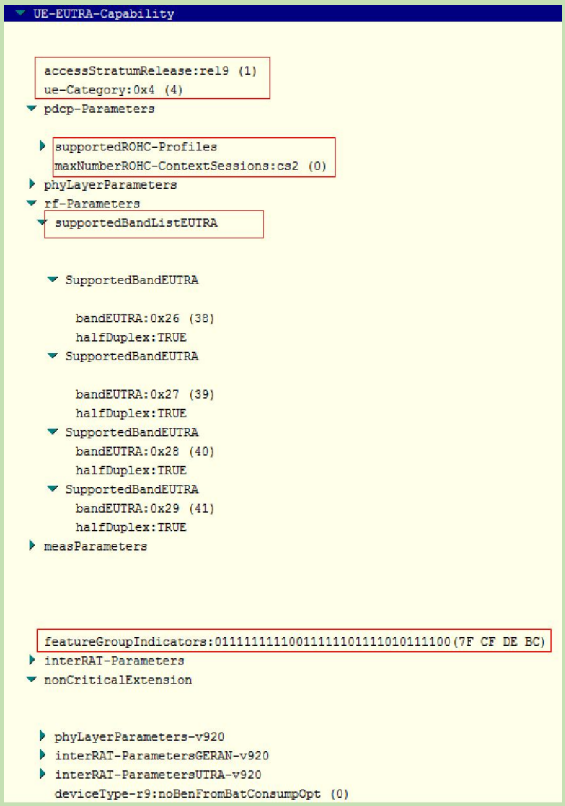
Idle 态 VoLTE UE 退出 idle 态时，同时建立 QCI9 和 QCI5 两个缺省承载。

若 UE 和 MME 都支持 SRVCC 能力，会携带 SRVCC Operation Possible IE。

通常，成功 attach 的终端，退出 idle 过程中，MME 会携带 UE Radio Capability IE，从该 IE 中，

可以获取到 UE 无线网络能力，如：RoHC 能力，FGI（SRVCC/SPS/TTI bundling 等能力）。

关键 IE	IE 取值定义
e-RABToBeSetupListCtxtSUReq SEQUENCE id:0x34 (52) criticality:reject (0) value e-RABToBeSetupItemCtxtSUReq e-RAB-ID:0x6 (6) e-RABlevelQoSParameters qCI:0x5 (5) allocationRetentionPriority priorityLevel:0x1 (1): highest pre-emptionCapability:shall-not-trigger-pre-emption (0) pre-emptionVulnerability:pre-emptable (1) transportLayerAddress:01100100010000011111011110010(64 41 FB D2) gTP-TEID:0F 9E 6A DE SEQUENCE id:0x34 (52) criticality:reject (0) value e-RABToBeSetupItemCtxtSUReq e-RAB-ID:0x5 (5) e-RABlevelQoSParameters qCI:0x9 (9) allocationRetentionPriority priorityLevel:0x6 (6) pre-emptionCapability:shall-not-trigger-pre-emption (0) pre-emptionVulnerability:pre-emptable (1) transportLayerAddress:01100100010000011111011110010(64 41 FB D3) gTP-TEID:25 EA D5 10 【解析】E-RAB to Be Setup Item IE 用来指示要建立 E-RAB 承载相关信息，包括 E-RAB ID，QoS（QCI，ARP 等），GTP-TEID，Transport Layer Address 信息等	36.413 9.1.4.1 INITIAL CONTEXT SETUP REQUEST E-RAB to Be Setup List | | 1 | | | >E-RAB to Be Setup Item IEs | | 1 .. <maxnoofE-RABs> | | | >>E-RAB ID | M | | 9.2.1.2 | >>E-RAB Level QoS Parameters | M | | 9.2.1.15 | Includes necessary QoS parameters. >>Transport Layer Address | M | | 9.2.2.1 | >>GTP-TEID | M | | 9.2.2.2 | >>NAS-PDU | O | | 9.2.3.5 | >>Correlation ID | O | | 9.2.1.80 | 9.2.1.15 E-RAB Level QoS Parameters IE/Group Name | Presence | Range | IE type and reference E-RAB Level QoS Parameters | | | | >QCI | M | | INTEGER (0..255) >Allocation and Retention Priority | M | | 9.2.1.60 >GBR QoS Information | O | | 9.2.1.18
sRVCCOperationPossible:possible (0) 【解析】该 IE 表示 UE 和 MME 支持 SRVCC	36.413 9.1.4.1 INITIAL CONTEXT SETUP REQUEST SRVCC Operation Possible | O | | 9.2.1.58 9.2.1.58 SRVCC Operation Possible This element indicates that both UE and MME are SRVCC-capable. E-UTRAN behaviour on receipt of this IE is specified in TS 23.216 [9]. IE/Group Name | Presence | Range | IE type and reference | Semantics description SRVCC operation possible | M | | ENUMERATED (Possible, ...) |



【解析】携带 UE 无线网络能力，该 IE 字段丰富，包括终端 RoHC 能力，SPS 能力，TTI bundling 能力，RLC UM mode 能力，SRVCC 能力等等

36.331 UE-EUTRA-Capability

36..331 B.1 Feature group indicators

该 IE 定义了 UE 在 E-UTRA 网络的无线能力。

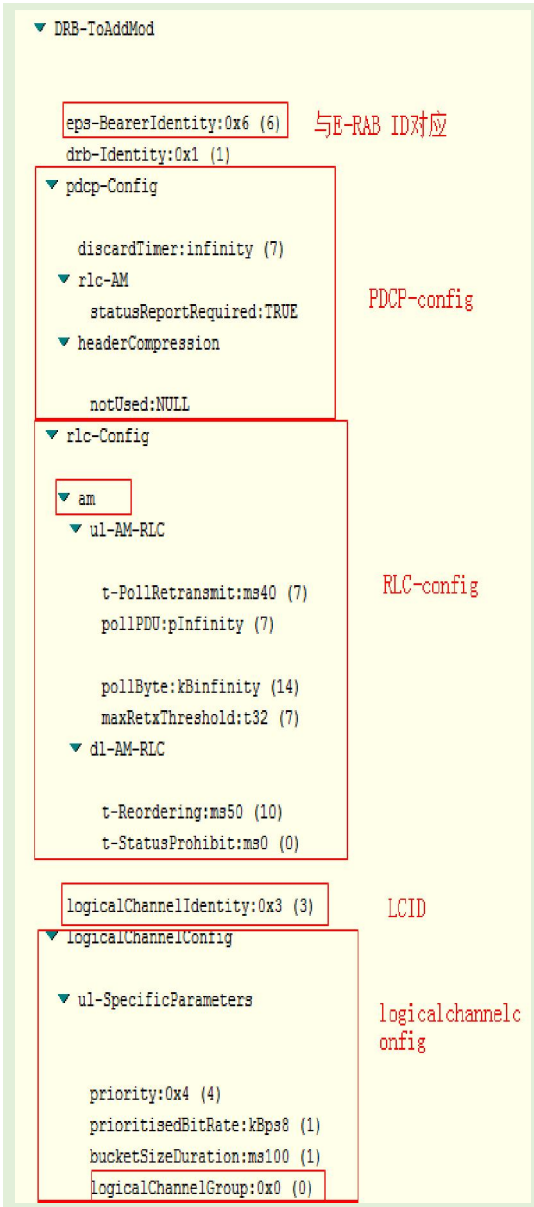
3.1.6 RRCConnectionReconfiguration（DRB setup-QCI9/QCI5）

收到核心网 MME 下发的 InitialContextSetupRequest 消息，根据 E-RAB to Be Setup Item 触发空口建立相应的 DRB（完成 PDCP-config，RLC-config，LogicalChannelConfig）。

关键 IE	IE 取值定义

	36.331 RadioResourceConfigDedicated
	DRB-ToAddMod ::= SEQUENCE {
	eps-BearerIdentity INTEGER (0..15) OPTIONAL, -- Cond DRB-Setup
	drb-Identity DRB-Identity,
	pdcp-Config PDCP-Config OPTIONAL, -- Cond PDCP
	rlc-Config RLC-Config OPTIONAL, -- Cond SetupM
	logicalChannelIdentity INTEGER (3..10) OPTIONAL, -- Cond DRB-Setup
	logicalChannelConfig LogicalChannelConfig OPTIONAL, -- Cond SetupM
	...,
	[[drb-TypeChange-r12 ENUMERATED {toMCG} OPTIONAL, -- Cond ToMCG
	rlc-Config-v12xy RLC-Config-v12xy OPTIONAL -- Cond Setup
	]]
	}

	36.331 PDCP-Config




```
PDCP-Config ::=                 SEQUENCE {
    discardTimer                    ENUMERATED {
                                        ms50, ms100, ms150, ms300, ms500,
                                        ms750, ms1500, infinity
    }                                                       OPTIONAL,       -- Cond Setup
    rlc-AM                          SEQUENCE {
        statusReportRequired            BOOLEAN
    }                                                       OPTIONAL,       -- Cond Rlc-AM
    rlc-UM                          SEQUENCE {
        pdcp-SN-Size                    ENUMERATED {len7bits, len12bits}
    }                                                       OPTIONAL,       -- Cond Rlc-UM
    headerCompression               CHOICE {
        notUsed                         NULL,
        rohc                            SEQUENCE {
            maxCID                          INTEGER (1..16383)          DEFAULT 15,
            profiles                        SEQUENCE {
                profile0x0001                   BOOLEAN,
                profile0x0002                   BOOLEAN,
                profile0x0003                   BOOLEAN,
                profile0x0004                   BOOLEAN,
                profile0x0006                   BOOLEAN,
                profile0x0101                   BOOLEAN,
                profile0x0102                   BOOLEAN,
                profile0x0103                   BOOLEAN,
                profile0x0104                   BOOLEAN
            },
            ...
        }
    },
    ...,
    [[  rn-IntegrityProtection-r10      ENUMERATED {enabled}    OPTIONAL    -- Cond RN
    ]],
    [[  pdcp-SN-Size-v1130              ENUMERATED {len15bits}  OPTIONAL    -- Cond Rlc-AM2
    ]],
    [[  ul-DataPath-r12                 ENUMERATED {rncg, scg}      OPTIONAL,   -- Need ON
        t-Reordering-r12                ENUMERATED {
                                            ms0, ms20, ms40, ms60, ms80, ms100, ms120, ms140,
                                            ms160, ms180, ms200, ms220, ms240, ms260, ms280, ms300,
                                            ms500, ms750, spare14, spare13, spare12, spare11, spare10,
                                            spare9, spare8, spare7, spare6, spare5, spare4, spare3,
                                            spare2, spare1}             OPTIONAL    -- Cond SetupS
    ]]
}
```

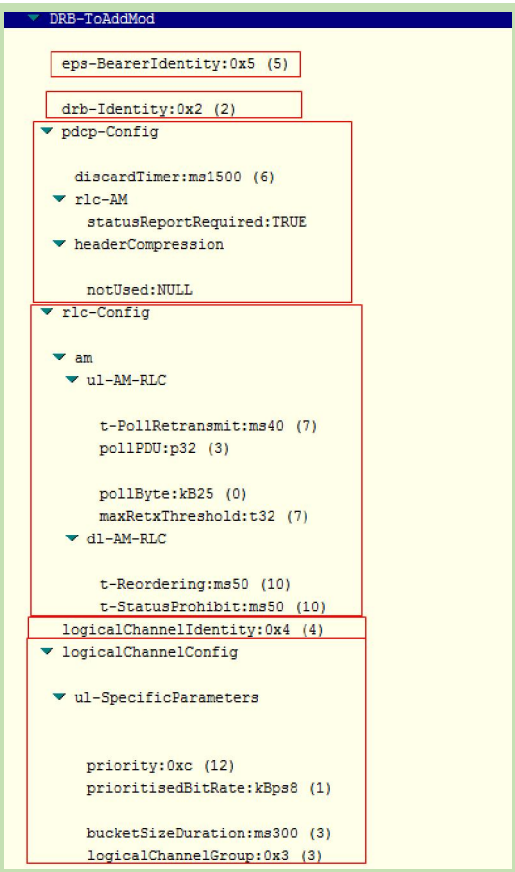
36.331 RLC-Config

```
RLC-Config ::=              CHOICE {
    am                                  SEQUENCE {
        ul-AM-RLC                           UL-AM-RLC,
        dl-AM-RLC                           DL-AM-RLC
    },
    um-Bi-Directional                   SEQUENCE {
        ul-UM-RLC                           UL-UM-RLC,
        dl-UM-RLC                           DL-UM-RLC
    },
    um-Uni-Directional-UL               SEQUENCE {
        ul-UM-RLC                           UL-UM-RLC
    },
    um-Uni-Directional-DL               SEQUENCE {
        dl-UM-RLC                           DL-UM-RLC
    },
    ...
}
```

```
RLC-Config-v12xy ::=                SEQUENCE {
    ul-extended-RLC-LI-Field-r12            BOOLEAN,
    dl-extended-RLC-LI-Field-r12            BOOLEAN
}
```

```
UL-AM-RLC ::=                   SEQUENCE {
    t-PollRetransmit                T-PollRetransmit,
    pollPDU                         PollPDU,
    pollByte                        PollByte,
    maxRetxThreshold                ENUMERATED {
                                        t1, t2, t3, t4, t6, t8, t16, t32}
}

DL-AM-RLC ::=                   SEQUENCE {
    t-Reordering                    T-Reordering,
    t-StatusProhibit                T-StatusProhibit
}
```



【解析】

（1）eps-BearerIdentity（即：E-RAB to Be Setup Item IE→E-RAB ID）和 drb-Identity 一一对应

（2）logicalChannelIdentity：逻辑信道 ID，MAC subheader 中包含该字段，用于指示相应的 MAC SDU 属于哪个逻辑信道

```
UL-UM-RLC ::=                   SEQUENCE {
    sn-FieldLength                  SN-FieldLength
}

DL-UM-RLC ::=                   SEQUENCE {
    sn-FieldLength                  SN-FieldLength,
    t-Reordering                    T-Reordering
}

SN-FieldLength ::=              ENUMERATED {size5, size10}
```

```
T-PollRetransmit ::=            ENUMERATED {
                                    ms5, ms10, ms15, ms20, ms25, ms30, ms35,
                                    ms40, ms45, ms50, ms55, ms60, ms65, ms70,
                                    ms75, ms80, ms85, ms90, ms95, ms100, ms105,
                                    ms110, ms115, ms120, ms125, ms130, ms135,
                                    ms140, ms145, ms150, ms155, ms160, ms165,
                                    ms170, ms175, ms180, ms185, ms190, ms195,
                                    ms200, ms205, ms210, ms215, ms220, ms225,
                                    ms230, ms235, ms240, ms245, ms250, ms300,
                                    ms350, ms400, ms450, ms500, spare9, spare8,
                                    spare7, spare6, spare5, spare4, spare3,
                                    spare2, spare1}

PollPDU ::=                     ENUMERATED {
                                    p4, p8, p16, p32, p64, p128, p256, pInfinity}

PollByte ::=                    ENUMERATED {
                                    kB25, kB50, kB75, kB100, kB125, kB250, kB375,
                                    kB500, kB750, kB1000, kB1250, kB1500, kB2000,
                                    kB3000, kBinfinity, spare1}

T-Reordering ::=                ENUMERATED {
                                    ms0, ms5, ms10, ms15, ms20, ms25, ms30, ms35,
                                    ms40, ms45, ms50, ms55, ms60, ms65, ms70,
                                    ms75, ms80, ms85, ms90, ms95, ms100, ms110,
                                    ms120, ms130, ms140, ms150, ms160, ms170,
                                    ms180, ms190, ms200, spare1}

T-StatusProhibit ::=            ENUMERATED {
                                    ms0, ms5, ms10, ms15, ms20, ms25, ms30, ms35,
                                    ms40, ms45, ms50, ms55, ms60, ms65, ms70,
                                    ms75, ms80, ms85, ms90, ms95, ms100, ms105,
                                    ms110, ms115, ms120, ms125, ms130, ms135,
                                    ms140, ms145, ms150, ms155, ms160, ms165,
                                    ms170, ms175, ms180, ms185, ms190, ms195,
                                    ms200, ms205, ms210, ms215, ms220, ms225,
                                    ms230, ms235, ms240, ms245, ms250, ms300,
                                    ms350, ms400, ms450, ms500, spare8, spare7,
                                    spare6, spare5, spare4, spare3, spare2,
                                    spare1}
```

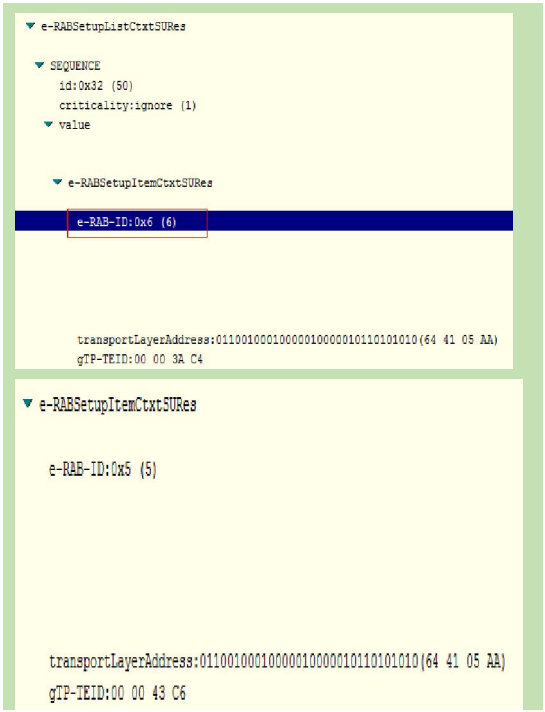
36.331 LogicalChannelConfig

```
LogicalChannelConfig ::=        SEQUENCE {
    ul-SpecificParameters           SEQUENCE {
        priority                        INTEGER (1..16),
        prioritisedBitRate              ENUMERATED {
                                            kBps0, kBps8, kBps16, kBps32, kBps64, kBps128,
                                            kBps256, infinity, kBps512-v1020, kBps1024-v1020,
                                            kBps2048-v1020, spare5, spare4, spare3, spare2,
                                            spare1},
        bucketSizeDuration              ENUMERATED {
                                            ms50, ms100, ms150, ms300, ms500, ms1000, spare2,
                                            spare1},
        logicalChannelGroup             INTEGER (0..3)          OPTIONAL            -- Need OR
    }   OPTIONAL,                                                                   -- Cond UL
    ...,
    [[  logicalChannelSR-Mask-r9        ENUMERATED {setup}      OPTIONAL        -- Cond SRmask
    ]],
    [[  logicalChannelSR-Prohibit-r12   BOOLEAN                 OPTIONAL        -- Need ON
    ]]
}
```

3.1.7 INITIAL CONTEXT SETUP RESPONSE

eNodeB 给 MME 回复 INITIAL CONTEXT SETUP RESPONSE 消息，确认 E-RAB 承载建立结果。

关键 IE	IE 取值定义



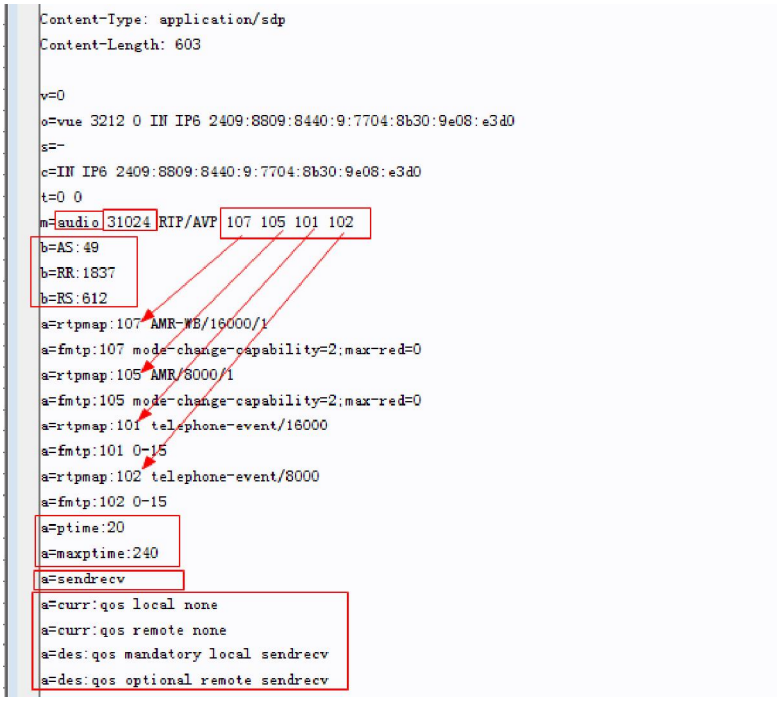
36.413 9.1.4.3 INITIAL CONTEXT SETUP RESPONSE

IE/Group Name	Presence	Range	IE type and reference
Message Type	M		9.2.1.1
MME UE S1AP ID	M		9.2.3.3
eNB UE S1AP ID	M		9.2.3.4
E-RAB Setup List		*1*	
>E-RAB Setup Item Ies		*1 .. <maxnoofE-RABs>*	
>>E-RAB ID	M		9.2.1.2
>>Transport Layer Address	M		9.2.2.1
>>GTP-TEID	M		9.2.2.2
E-RAB Failed to Setup List	O		E-RAB List 9.2.1.36

3.1.8 SIP INVITE

用户发起 VoLTE 呼叫，UE 构造 SIP INVITE 消息（携带 SDP offer 信息），在 UE 进入 Active 态后，UE 在 QCI5 default EPS bearer 上将 SIP INVITE 发送给 P-CSCF。

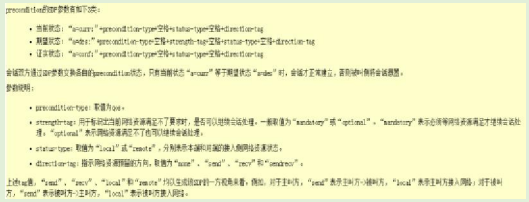




关键 IE	IE 取值定义
Request URI sip:17820500319;phone-context=gd.ims.mnc000.mcc460.3gppnetwork.org@gd.ims.mnc000.mcc460.3gppnetwork.org;user=phone SIP/2.0 【解析】用于标明请求的目的方，可以为 SIP URI 和 tel URI	RFC 3261 7.1 Requests Request-URI: The Request-URI is a SIP or SIPS URI as described in Section 19.1 or a general URI (RFC 2396 [5]). It indicates the user or service to which this request is being addressed. The Request-URI MUST NOT contain unescaped spaces or control characters and MUST NOT be enclosed in "<>". SIP elements MAY support Request-URIs with schemes other than "sip" and "sips", for example the "tel" URI scheme of RFC 2806 [9]. SIP elements MAY translate non-SIP URIs using any mechanism at their disposal, resulting in SIP URI, SIPS URI, or some other scheme. 19.1 SIP and SIPS Uniform Resource Indicators RFC 3966 The tel URI for Telephone Numbers
From: <sip:+8617820500317@gd.ims.mnc000.mcc460.3gppnetwork.org>;tag=rrfcbjy 【解析】用于标明请求的发起方，携带请求发起者的 URI	RFC 3261： 20.20 From The From header field indicates the initiator of the request. This may be different from the initiator of the dialog. Requests sent by the callee to the caller use the callee's address in the From header field. The optional "display-name" is meant to be rendered by a human user interface. A system SHOULD use the display name "Anonymous" if the identity of the client is to remain hidden. Even if the "display-name" is empty, the "name-addr" form MUST be used if the "addr-spec" contains a comma, question mark, or semicolon. Syntax issues are discussed in Section 7.3.1. Two From header fields are equivalent if their URIs match, and their parameters match. Extension parameters in one header field, not present in the other are ignored for the purposes of comparison. This means that the display name and presence or absence of angle brackets do not affect matching. See Section 20.10 for the rules for parsing a display name, URI and URI parameters, and header field parameters. The compact form of the From header field is f. Examples: From: "A. G. Bell" <sip:agb@bell-telephone.com> ;tag=a48s From: sip:+12125551212@server.phone2net.com;tag=887s f: Anonymous <sip:c8oqz84zk7z@privacy.org>;tag=hyh8 19.3 Tags The "tag" parameter is used in the To and From header fields of SIP messages. It serves as a general mechanism to identify a dialog, which is the combination of the Call-ID along with two tags, one from each participant in the dialog. When a UA sends a request outside of

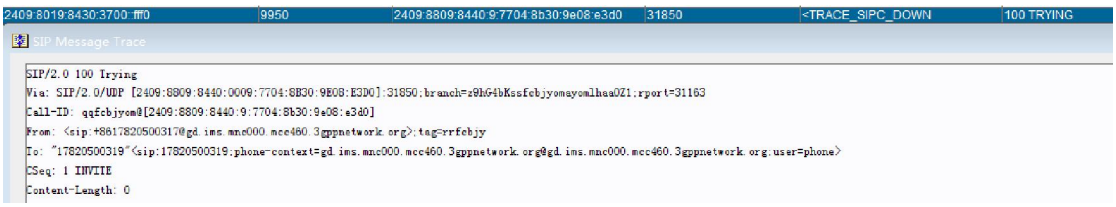
To: "17820500319"<sip:17820500319;phone-context=gd.ims.mnc000.mcc460.3gppnetwork.org@gd.ims.mnc000.mcc460.3gppnetwork.org;user=phone> 【解析】用于标明请求的接收方，携带请求接收者的 URI。	RFC 3261： 20.39 To The To header field specifies the logical recipient of the request. The optional "display-name" is meant to be rendered by a human-user interface. The "tag" parameter serves as a general mechanism for dialog identification. See Section 19.3 for details of the "tag" parameter. Comparison of To header fields for equality is identical to comparison of From header fields. See Section 20.10 for the rules for parsing a display name, URI and URI parameters, and header field parameters. The compact form of the To header field is t. The following are examples of valid To header fields: To: The Operator <sip:operator@cs.columbia.edu>;tag=287447 t: sip:+12125551212@server.phone2net.com
Contact: <sip:460075205000317@[2409:8809:8440:9:7704:8b30:9e08:e3d0]:31850>;+sip.instance="<urn:gsma:imei:86516602-493704-0>";+g.3gpp.icsi-ref="urn%3Aurn-7%3A3gpp-service.ims.icsi.mmtel";+g.3gpp.mid-call;+g.3gpp.srvcc-alerting 【解析】用于标明直接联系请求发送方或应答方的 URI 地址 +g.3gpp.mid-call：表示 UE 支持 mid-call SRVCC 切换 +g.3gpp.srvcc-alerting：表示 UE 支持 Alerting SRVCC 切换	24.237 C.2 Definition of media feature tag g.3gpp.mid-call Media feature-tag name: g.3gpp.mid-call ASN.1 Identifier: 1.3.6.1.8.2.12 Summary of the media feature indicated by this tag: This feature-tag when used in a SIP request or a SIP response indicates that the function sending the SIP message supports the MSC server assisted mid-call feature. Values appropriate for use with this feature-tag: Boolean 24.237 C.5 Definition of media feature tag g.3gpp.srvcc-alerting Media feature-tag name: g.3gpp.srvcc-alerting Summary of the media feature indicated by this tag: This media feature-tag when used in a Contact header field of a SIP request or a SIP response indicates that the functional entity sending the SIP message supports the PS to CS SRVCC for calls in alerting phase, i.e. for calls with early dialog. Values appropriate for use with this feature-tag: Boolean
P-Access-Network-Info: 3GPP-E-UTRAN-TDD;utran-cell-id-3gpp=4600027912C70801 【解析】PANI 头域应该携带终端当前接入网络的信息	24.229 7.2A.4 P-Access-Network-Info header field 7.2A.4.1 Introduction The P-Access-Network-Info header field is extended to include specific information relating to particular access technologies. RFC 3455 Private Header (P-Header) Extensions to the Session Initiation Protocol (SIP) for the 3rd-Generation Partnership Project (3GPP)
m=audio 31024 RTP/AVP 107 105 101 102 a=rtpmap:107 AMR-WB/16000/1 a=fmtp:107 mode-change-capability=2;max-red=0 a=rtpmap:105 AMR/8000/1 a=fmtp:105 mode-change-capability=2;max-red=0	RFC 4566 SDP: Session Description Protocol 5.14. Media Descriptions ("m=") 5.13. Attributes ("a=") 6. SDP Attributes

a=rtpmap:101 telephone-event/16000 a=fmtp:101 0-15 a=rtpmap:102 telephone-event/8000 a=fmtp:102 0-15 【解析】SDP 中携带主叫支持的 codec 类型，媒体类型等信息，如：m=audio 31024 RTP/AVP 107 105 101 102，这里表示媒体类型为 audio，媒体包 RTP 传输端口号为 31024（隐含表示 RTCP 传输端口号为 31025），传输协议为 RTP/AVP，支持的媒体 payload type number 为 107 105 101 102，后面相应的 a 行定义 payload type number 所对应的编解码类型信息。	RFC 3550 11. RTP over Network and Transport Protocols RTP relies on the underlying protocol(s) to provide demultiplexing of RTP data and RTCP control streams. For UDP and similar protocols, RTP SHOULD use an even destination port number and the corresponding RTCP stream SHOULD use the next higher (odd) destination port number.
b=AS:49 b=RR:1837 b=RS:612 【解析】用来指示 RTP/RTCP 所需带宽	RFC 4566 SDP: Session Description Protocol **5.8. Bandwidth ("b=")** b=<bwtype>:<bandwidth> AS The bandwidth is interpreted to be application specific (it will be the application's concept of maximum bandwidth). Normally, this will coincide with what is set on the application's "maximum bandwidth" control if applicable. For RTP-based applications, AS gives the RTP "session bandwidth" as defined in Section 6.2 of [19]. The <bandwidth> is interpreted as kilobits per second by default. RFC 3556 Session Description Protocol (SDP) Bandwidth Modifiers for RTP Control Protocol (RTCP) Bandwidth b=RS:<bandwidth-value> b=RR:<bandwidth-value> The <bandwidth-value> for these modifiers is in units of bits per second with an integer value.
Supported: 100rel,histinfo,join,norefersub,precondition,replaces,timer,sec-agree …	RFC 3312 Integration of Resource Management and Session Initiation Protocol (SIP) 24.229

a=curr:qos local none a=curr:qos remote none a=des:qos mandatory local sendrecv a=des:qos optional remote sendrecv 【解析】Supported 头域中指示本端支持 precondition，同时 SDP 中携带 precondition 相关的 QoS 参数。 	5.1.3 Call initiation - UE-originating case 5.1.3.1 Initial INVITE request 规定了主叫 UE 对 Precondition 机制的要求
a=ptime:20 a=maxptime:240 【解析】语音包打包时长信息	RFC 4566 SDP: Session Description Protocol **6. SDP Attributes** a=ptime:<packet time> This gives the length of time in milliseconds represented by the media in a packet. This is probably only meaningful for audio data, but may be used with other media types if it makes sense. It should not be necessary to know ptime to decode RTP or vat audio, and it is intended as a recommendation for the encoding/packetisation of audio. It is a media-level attribute, and it is not dependent on charset. a=maxptime:<maximum packet time> This gives the maximum amount of media that can be encapsulated in each packet, expressed as time in milliseconds. The time SHALL be calculated as the sum of the time the media present in the packet represents. For frame-based codecs, the time SHOULD be an integer multiple of the frame size. This attribute is probably only meaningful for audio data, but may be used with other media types if it makes sense. It is a media-level attribute, and it is not dependent on charset. Note that this attribute was introduced after RFC 2327, and non-updated implementations will ignore this attribute.

3.1.9 SIP 100 TRYING

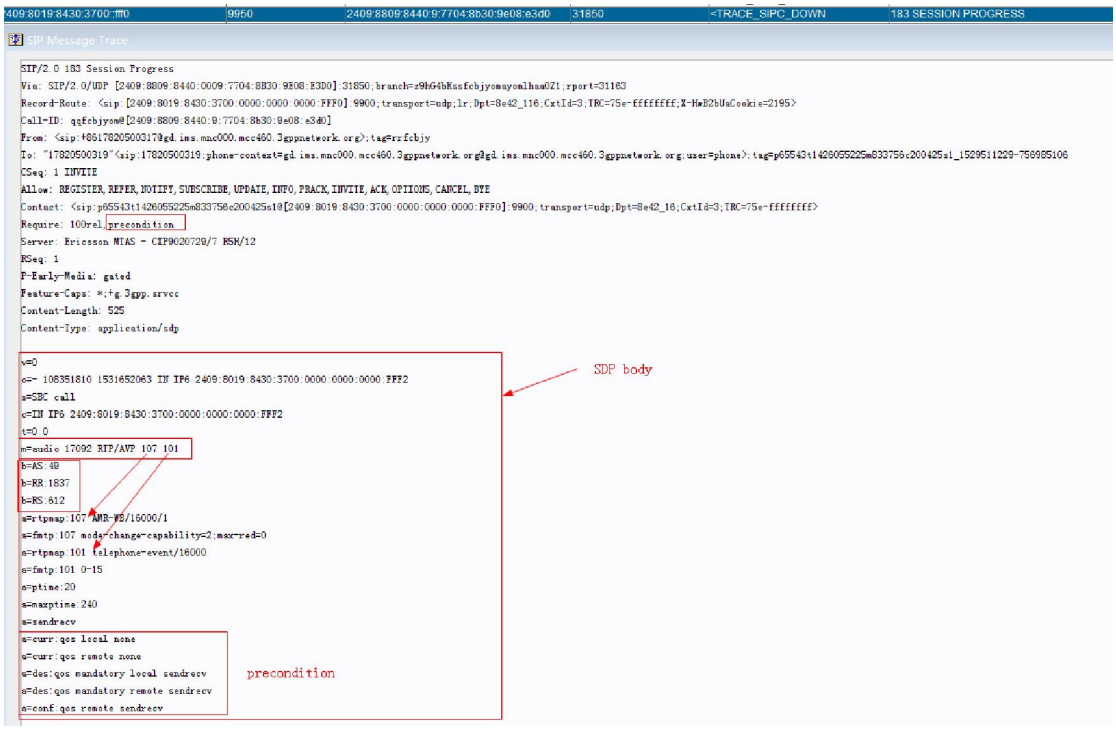
P-CSCF 收到 INVITE 消息后，回复临时响应消息 100 TRYING 给 UE，用以停止 UE 重发 INVITE 消息。



3.1.10 SIP 183 Session Progress

被叫终端回复 183 Session Progress（携带 SDP Answer），经过 IMS 核心网设备转发给主叫用户接入的 P-CSCF。

如下 183 session progress 消息为主叫 P-CSCF 发送给主叫 UE 信令截图：



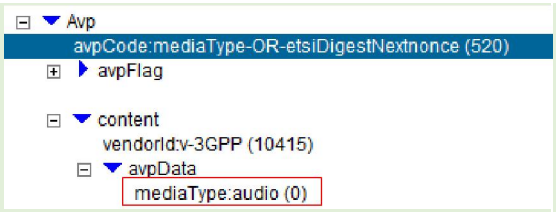
此后，P-CSCF 根据 INVITE（SDP Offer）和 183 Session Progress（SDP Answer）协商 SDP 结果，发送 AAR 消息给 PCRF，触发 EPS 专用承载（QCI1）建立。

该 183 Session Progress 表示协商的 codec 为 AMR-WB，相关 IE 定义与 SIP INVITE 相同。

3.1.11 AAR

主叫 P-CSCF 收到被叫 183 Session Progress（SDP Answer）后，触发 AAR 消息给 PCRF，触发专用承载（QCI1）建立，携带媒体业务流相关信息，媒体带宽需求等信息。





【解析】指示媒体类型

29.214 5.3.19 Media-Type AVP

The Media-Type AVP (AVP code 520) is of type Enumerated, and it determines the media type of a session component. The media types indicate the type of media in the same way as the SDP media types with the same names defined in RFC 4566 [13]. The following values are defined:

- AUDIO (0)
- VIDEO (1)
- DATA (2)
- APPLICATION (3)
- CONTROL (4)
- TEXT (5)
- MESSAGE (6)
- OTHER (0xFFFFFFFF)



【解析】下行 IP flow 所需要最大带宽

29.214 5.3.14 Max-Requested-Bandwidth-DL AVP

The Max-Requested-Bandwidth-DL AVP (AVP code 515) is of type Unsigned32, and it indicates the maximum bandwidth in bits per second for a downlink IP flow. The bandwidth contains all the overhead coming from the IP-layer and the layers above, e.g. IP, UDP, RTP and RTP payload.

When provided in an AA-Request, it indicates the maximum requested bandwidth. When provided in an AA-Answer, it indicates the maximum bandwidth acceptable by PCRF.

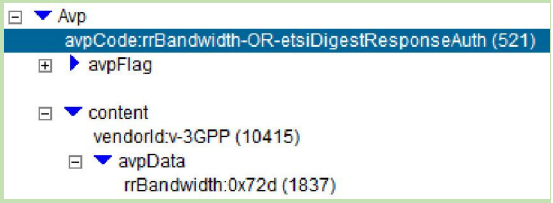


【解析】上行 IP flow 所需要最大带宽

29.214 5.3.15 Max-Requested-Bandwidth-UL AVP

The Max –Bandwidth-UL AVP (AVP code 516) is of type Unsigned32, and it indicates the maximum requested bandwidth in bits per second for an uplink IP flow. The bandwidth contains all the overhead coming from the IP-layer and the layers above, e.g. IP, UDP, RTP and RTP payload.

When provided in an AA-Request, it indicates the maximum requested bandwidth. When provided in an AA-Answer, it indicates the maximum bandwidth acceptable by PCRF.



【解析】指示 RTCP receiver report 所需要的带宽

29.214 5.3.20 RR-Bandwidth AVP

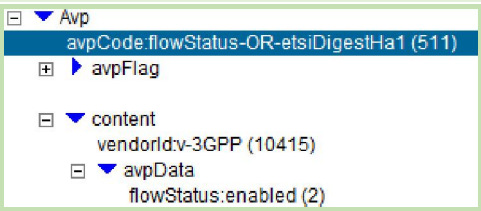
The RR-Bandwidth AVP (AVP code 521) is of type Unsigned32, and it indicates the maximum required bandwidth in bits per second for RTCP receiver reports within the session component, as specified in RFC 3556 [11]. The bandwidth contains all the overhead coming from the IP-layer and the layers above, i.e. IP, UDP and RTCP.



【解析】指示 RTCP sender report 所需要的带宽

29.214 5.3.21 RS-Bandwidth AVP

The RS-Bandwidth AVP (AVP code 522) is of type Unsigned32, and it indicates the maximum required bandwidth in bits per second for RTCP sender reports within the session component, as specified in RFC 3556 [11]. The bandwidth contains all the overhead coming from the IP-layer and the layers above, i.e. IP, UDP and RTCP.



29.214 5.3.11 Flow-Status AVP

【解析】指示该 IP flow 是否 enable，enable 状态会放通 IP flow 上的媒体流数据包

The Flow-Status AVP (AVP code 511) is of type Enumerated, and describes whether the IP flow(s) are enabled or disabled. The following values are defined:

ENABLED-UPLINK (0)

This value shall be used to enable associated uplink IP flow(s) and to disable associated downlink IP flow(s).

ENABLED-DOWNLINK (1)

This value shall be used to enable associated downlink IP flow(s) and to disable associated uplink IP flow(s).

ENABLED (2)

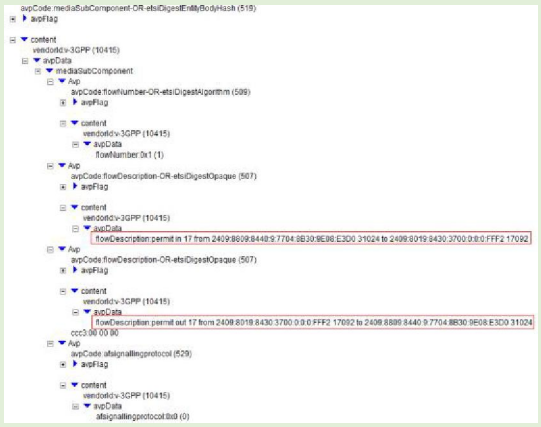
This value shall be used to enable all associated IP flow(s) in both directions.

DISABLED (3)

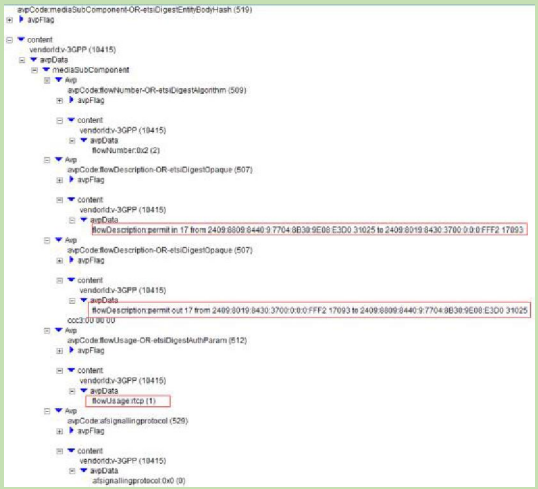
This value shall be used to disable all associated IP flow(s) in both directions.

REMOVED (4)

This value shall be used to remove all associated IP flow(s). The IP Filters for the associated IP flow(s) shall be removed. The associated IP flows shall not be taken into account when deriving the authorized QoS.



【解析】RTP 业务流 packet filter 信息



【解析】RTCP 业务流 packet filter 信息

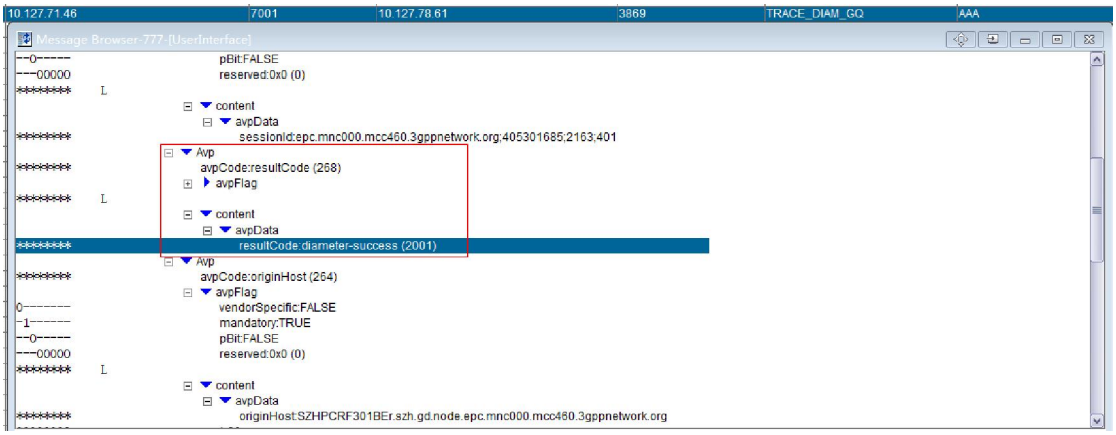
24.229 5.3.8 Flow-Description AVP

5.3.9 Flow-Number AVP

5.3.12 Flow-Usage AVP

3.1.12 AAA

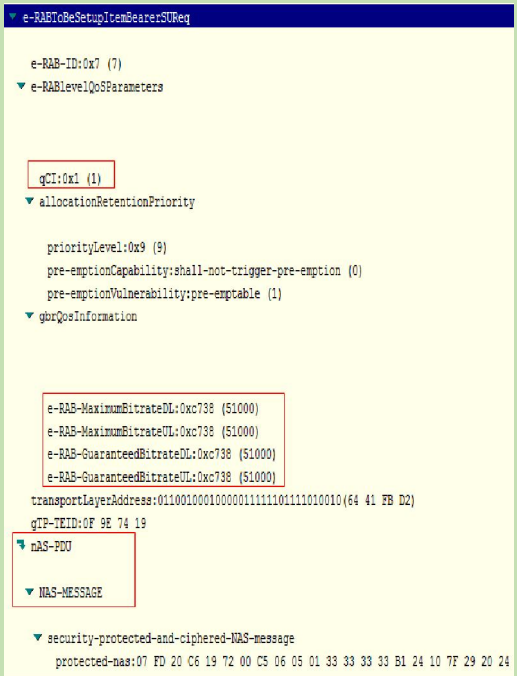
PCRF 发送 AAA 消息对 AAR 消息进行应答。



3.1.13 E-RAB SETUP REQUEST

MME 发送 E-RAB SETUP REQUEST 消息，请求建立 E-RAB 承载（QCI1），用来承载 VOLTE 媒体面数据包。

关键 IE	IE 取值定义



【解析】MME 触发建立 E-RAB 承载（QCI1），GBR=MBR=51kbp，相应的携带 NAS PDU 信元，由 eNodeB 转发给 UE

36.413 9.1.3.1 E-RAB SETUP REQUEST

E-RAB to be Setup List		1		
>E-RAB To Be Setup Item IEs		1 .. <maxnoof E-RABs>		
>>E-RAB ID	M		9.2.1.2	
>>E-RAB Level QoS Parameters	M		9.2.1.15	Includes necessary QoS parameters.
>>Transport Layer Address	M		9.2.2.1	
>>GTP-TEID	M		9.2.2.2	EPC TEID.
>>NAS-PDU	M		9.2.3.5	

9.2.1.15 E-RAB Level QoS Parameters

IE/Group Name	Presence	Range	IE type and reference	Semantics description
E-RAB Level QoS Parameters				
>QCI	M		INTEGER (0..255)	QoS Class Identifier defined in TS 23.401 [11]. Coding specified in TS 23.203 [13].
>Allocation and Retention Priority	M		9.2.1.60	
>GBR QoS Information	O		9.2.1.18	This IE applies to GBR bearers only and shall be ignored otherwise.

9.2.1.18 GBR QoS Information

IE/Group Name	Presence	Range	IE type and reference	Semantics description
E-RAB Maximum Bit Rate Downlink	M		Bit Rate 9.2.1.19	**Desc.**: This IE indicates the maximum downlink E-RAB Bit Rate as specified in TS 23.401 [11] for this bearer.
E-RAB Maximum Bit Rate Uplink	M		Bit Rate 9.2.1.19	**Desc.**: This IE indicates the maximum uplink E-RAB Bit Rate as specified in TS 23.401 [11] for this bearer.
E-RAB Guaranteed Bit Rate Downlink	M		Bit Rate 9.2.1.19	**Desc.**: This IE indicates the downlink guaranteed E-RAB Bit Rate as specified in TS 23.401 [11] (provided that there is data to deliver) for this bearer.
E-RAB Guaranteed Bit Rate Uplink	M		Bit Rate 9.2.1.19	**Desc.**: This IE indicates the uplink guaranteed E-RAB Bit Rate as specified in TS 23.401 [11] (provided that there is data to deliver) for this bearer.

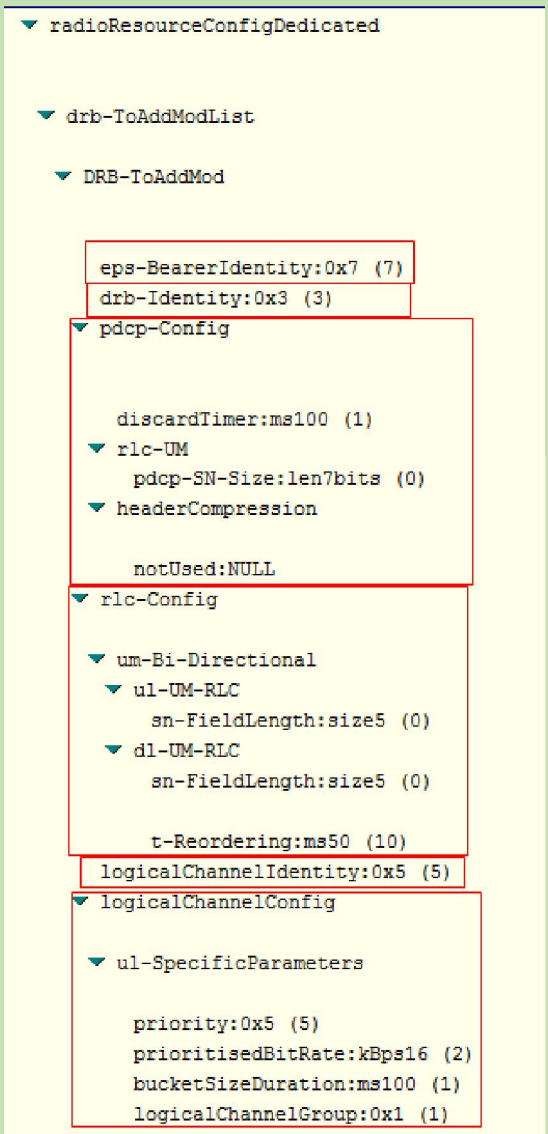
9.2.1.19 Bit Rate

IE/Group Name	Presence	Range	IE type and reference	Semantics description
Bit Rate			INTEGER (0..10,000,000,000)	The unit is: bit/s.

3.1.14 RRCConnectionReconfiguration（DRB setup-QCI1）

eNodeB 根据 MME 下发的 E-RAB SETUP REQUEST，触发建立空口 DRB 承载，同时，将 E-RAB SETUP REQUEST 消息中的 NAS PDU 通过 RRCConnectionReconfiguration→dedicatedInfoNASList。

关键 IE	IE 取值定义



参考 RRCConnectionReconfiguration （DRB setup-QCI9/QCI5）

3.1.15 ACTIVATE DEDICATED EPS BEARER CONTEXT REQUEST

专用承载建立时，核心网 MME 会下发 ACTIVATE DEDICATED EPS BEARER CONTEXT REQUEST，请求为已有缺省承载的 PDN connection 激活专有承载（这里为 QCI1）。

关键 IE	IE 取值定义
	24.301 8.3.3 Activate dedicated EPS bearer context request

```
▼ no-security-protection-SM-message
    ePS-bearer-identity:0x7 (7)        dedicated EPS bearer ID
    procedure-transaction-identity:0x0 (0)
  ▼ msg-body

    ▼ activateDedicatedEPSBearerContextRequest
        spare-half-octet:0x0 (0)
        linked-EPS-bearer-identity:ePS-bearer-identity-value6 (6)
```

【解析】新建 dedicated EPS bearer ID 为 7，关联的 default EPS bearer ID 为 6

IEI	Information Element	Type/Reference	Presence	Format	Length
	Protocol discriminator	Protocol discriminator 9.2	M	V	1/2
	EPS bearer identity	EPS bearer identity 9.3.2	M	V	1/2
	Procedure transaction identity	Procedure transaction identity 9.4	M	V	1
	Activate dedicated EPS bearer context request message identity	Message type 9.8	M	V	1
	Linked EPS bearer identity	Linked EPS bearer identity 9.9.4.6	M	V	1/2
	Spare half octet	Spare half octet 9.9.2.9	M	V	1/2
	EPS QoS	EPS quality of service 9.9.4.3	M	LV	2-14
	TFT	Traffic flow template 9.9.4.16	M	LV	2-256

9.3.2 EPS bearer identity

Bits 5 to 8 of the first octet of every EPS Session Management (ESM) message contain the EPS bearer identity. The EPS bearer identity and its use to identify a message flow are defined in 3GPP TS 24.007 [12].

9.9.4.6 Linked EPS bearer identity

The purpose of the Linked EPS bearer identity IE is to identify the default bearer that is associated with a dedicated EPS bearer or to identify the EPS bearer (default or dedicated) with which one or more packet filters specified in a traffic flow aggregate are associated.

```
▼ sDF-QoS
    qCI:0x1 (1)
    maximum-bit-rate-for-uplink:0x33 (51)
    maximum-bit-rate-for-downlink:0x33 (51)
    guaranteed-bit-rate-for-uplink:0x33 (51)
    guaranteed-bit-rate-for-downlink:0x33 (51)
```

【解析】dedicated EPS bearer QoS 参数（QCI1，MBR,GBR）

9.9.4.3 EPS quality of service

8	7	6	5	4	3	2	1	
EPS quality of service IEI								octet 1
Length of EPS quality of service contents								octet 2
QCI								octet 3
Maximum bit rate for uplink								octet 4*
Maximum bit rate for downlink								octet 5*
Guaranteed bit rate for uplink								octet 6*
Guaranteed bit rate for downlink								octet 7*
Maximum bit rate for uplink (extended)								octet 8*
Maximum bit rate for downlink (extended)								octet 9*
Guaranteed bit rate for uplink (extended)								octet 10*
Guaranteed bit rate for downlink (extended)								octet 11*
Maximum bit rate for uplink (extended-2)								octet 12*
Maximum bit rate for downlink (extended-2)								octet 13*
Guaranteed bit rate for uplink (extended-2)								octet 14*
Guaranteed bit rate for downlink (extended-2)								octet 15*

Figure 9.9.4.3.1: EPS quality of service information element

【注意】当 MBR 和 GBR 速率大于 64kbps 后，需要根据协议折算实际速率取值

```
.tFT
...tft-operation-code:create-new-TFT (1)   新建TFT
...e-bit:parameters-list-is-not-included (0)
...number-of-packet-filters:0x4 (4)
...packet-filter-list2
....Create-packet-filter   新建packet filter
.......spare:0x0 (0)
.......packet-filter-direction:downlink-only (1)
.......packet-filter-id:0x0 (0)
.......packet-filter-evaluation-precedence:0x7f (127)   packet filter

.......packet-filter-content
.........Packet-filter-content
...........packet-filter-component-type-id:iPv6-remote-addr (32)
...........packet-filter-content-value      remote address（网络侧地址）
.............ipv6

.............ipv6-addr:24 09 80 19 84 30 37 00 00 00 00 00 00 00 FF F1
.............ipv6-mask:FF FF FF FF FF FF FF FF FF FF FF FF FF FF FF FF
.........Packet-filter-content
...........packet-filter-component-type-id:protocol-identifier-Next-header (48)
...........packet-filter-content-value
.............protocol-id-or-Next-header-value:0x11 (17)  协议类型
.........Packet-filter-content
...........packet-filter-component-type-id:single-local-port (64)
...........packet-filter-content-value

.............single-local-port:0x7931 (31025)  local port（UE侧端口）
.........Packet-filter-content
...........packet-filter-component-type-id:single-remote-port (80)
...........packet-filter-content-value

.............single-remote-port:0x42C5 (17093) remote port（网络侧端口）
```

24.008 10.5.6.12 Traffic Flow Template

The *packet filter evaluation precedence* field is used to specify the precedence for the packet filter among all packet filters in all TFTs associated with this PDP address. Higher the value of the *packet filter evaluation precedence* field, lower the precedence of that packet filter is. The first bit in transmission order is the most

The term *local* refers to the MS and the term *remote* refers to an external network entity.

Packet filter component type identifier

Bits	
8 7 6 5 4 3 2 1	
0 0 0 1 0 0 0 0	IPv4 remote address type
0 0 0 1 0 0 0 1	IPv4 local address type
0 0 1 0 0 0 0 0	IPv6 remote address type
0 0 1 0 0 0 0 1	IPv6 remote address/prefix length type
0 0 1 0 0 0 1 1	IPv6 local address/prefix length type
0 0 1 1 0 0 0 0	Protocol identifier/Next header type
0 1 0 0 0 0 0 0	Single local port type
0 1 0 0 0 0 0 1	Local port range type
0 1 0 1 0 0 0 0	Single remote port type
0 1 0 1 0 0 0 1	Remote port range type
0 1 1 0 0 0 0 0	Security parameter index type
0 1 1 1 0 0 0 0	Type of service/Traffic class type
1 0 0 0 0 0 0 0	Flow label type

All other values are reserved.

```
.Create-packet-filter
...spare:0x0 (0)
...packet-filter-direction:uplink-only (2)
...packet-filter-id:0x1 (1)
...packet-filter-evaluation-precedence:0x80 (128)

...packet-filter-content
.....Packet-filter-content
.......packet-filter-component-type-id:iPv6-remote-addr (32)
.......packet-filter-content-value
.........ipv6
...........ipv6-addr:24 09 80 19 84 30 37 00 00 00 00 00 00 00 FF F1
...........ipv6 mask:FF FF FF FF FF FF FF FF FF FF FF FF FF FF FF FF
.....Packet-filter-content
.......packet-filter-component-type-id:protocol-identifier-Next-header (48)
.......packet-filter-content-value
.........protocol-id-or-Next-header-value:0x11 (17)
.....Packet-filter-content
.......packet-filter-component-type-id:single-local-port (64)
.......packet-filter-content-value

.........single-local-port:0x7931 (31025)
.....Packet-filter-content
.......packet-filter-component-type-id:single-remote-port (80)
.......packet-filter-content-value

.........single-remote-port:0x42C5 (17093)
```

```
..Create-packet-filter
....spare:0x0 (0)
....packet-filter-direction:downlink-only (1)
....packet-filter-id:0x2 (2)
....packet-filter-evaluation-precedence:0x3e (62)

....packet-filter-content
......Packet-filter-content
........packet-filter-component-type-id:iPv6-remote-addr (32)
........packet-filter-content-value
..........ipv6

............ipv6-addr:24 09 80 19 84 30 37 00 00 00 00 00 00 00 FF F1
............ipv6-mask:FF FF FF FF FF FF FF FF FF FF FF FF FF FF FF FF
......Packet-filter-content
........packet-filter-component-type-id:protocol-identifier-Next-header (48)
........packet-filter-content-value
..........protocol-id-or-Next-header-value:0x11 (17)
......Packet-filter-content
........packet-filter-component-type-id:single-local-port (64)
........packet-filter-content-value

..........single-local-port:0x7930 (31024)
......Packet-filter-content
........packet-filter-component-type-id:single-remote-port (80)
........packet-filter-content-value

..........single-remote-port:0x42C4 (17092)
```

```
.Create-packet-filter
...spare:0x0 (0)
...packet-filter-direction:uplink-only (2)
...packet-filter-id:0x3 (3)
...packet-filter-evaluation-precedence:0x3f (63)

...packet-filter-content
.....Packet-filter-content
.......packet-filter-component-type-id:iPv6-remote-addr (32)
.......packet-filter-content-value
.........ipv6
...........ipv6-addr:24 09 80 19 84 30 37 00 00 00 00 00 00 00 FF F1
...........ipv6-mask:FF FF FF FF FF FF FF FF FF FF FF FF FF FF FF FF
.....Packet-filter-content
.......packet-filter-component-type-id:protocol-identifier-Next-header (48)
.......packet-filter-content-value
.........protocol-id-or-Next-header-value:0x11 (17)
.....Packet-filter-content
.......packet-filter-component-type-id:single-local-port (64)
.......packet-filter-content-value

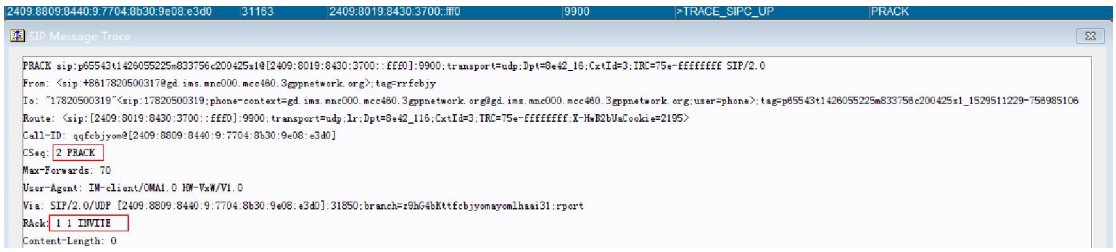
.........single-local-port:0x7930 (31024)
.....Packet-filter-content
.......packet-filter-component-type-id:single-remote-port (80)
.......packet-filter-content-value

.........single-remote-port:0x42C4 (17092)
```

【解析】创建 TFT-packet filter，用以将上层业务包映射到相应的 dedicated EPS bearer。

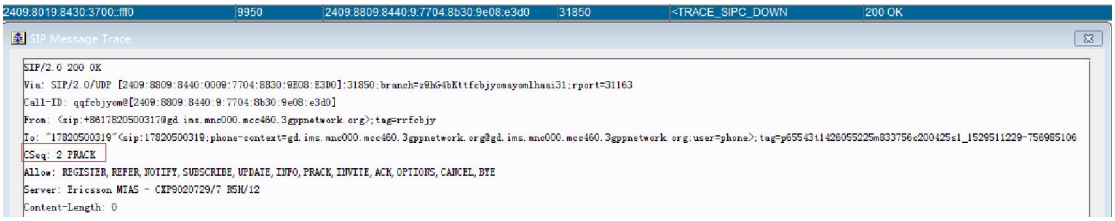
3.1.16 SIP PRACK

终端收到 SIP 183 session Progress，回复 PRACK 进行应答.



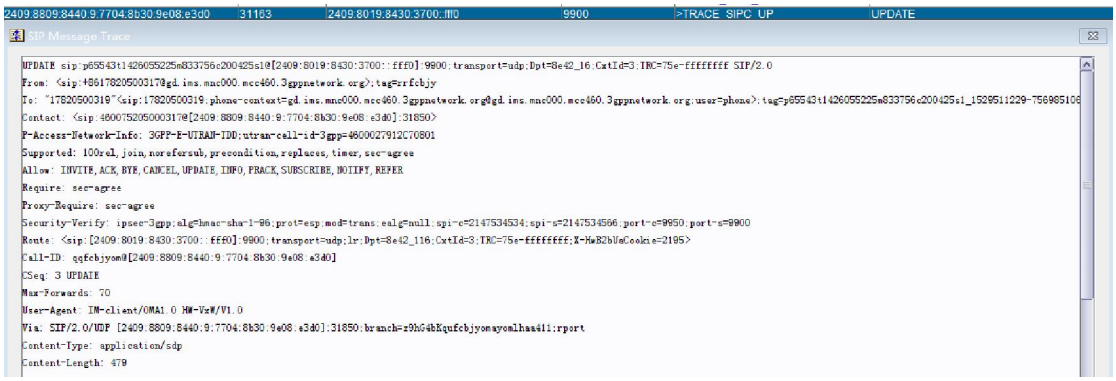
3.1.17 SIP 200 OK(PRACK)

被叫终端收到 PRACK 消息后，回复 SIP 200 OK 对 PRACK 进行确认，通过核心网发送给主叫 P-CSCF，再发给主叫 UE：



3.1.18 SIP UPDATE

主叫终端完成资源预留，即：QCI1 承载建立完成后，发送 SIP UPDATE 消息，指示本端 Precondition 已满足。



```
v=0
o=vue 3212 3212 IN IP6 2409:8809:8440:9:7704:8b30:9e08:e3d0
s=-
c=IN IP6 2409:8809:8440:9:7704:8b30:9e08:e3d0
t=0 0
m=audio 31024 RIP/AVP 107 101
b=AS:49
b=RR:1837
b=RS:612
a=rtpmap:107 AMR-WB/16000/1
a=fmtp:107 mode-change-capability=2;max-red=0
a=rtpmap:101 telephone-event/16000
a=fmtp:101 0-15
a=ptime:20
a=maxptime:240
a=sendrecv
a=curr:qos local sendrecv
a=curr:qos remote none
a=des:qos mandatory local sendrecv
a=des:qos mandatory remote sendrecv
```

本端Precondition条件已满足

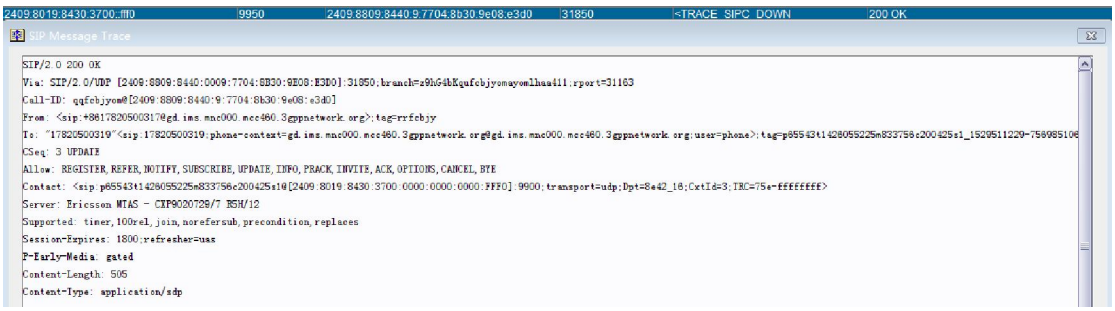
相关 IE 定义与 SIP INVITE 相同。

3.1.19 SIP 200 OK（UPDATE）

被叫侧完成资源预留后，返回 SIP 200 OK (UPDATE)给主叫侧，参数“a=curr:qos local sendrecv”指示被叫侧 Precondition 已经满足。

被叫侧 SIP 200 OK（UPDATE）经过核心网发送给主叫 P-CSCF，再发送给主叫 UE。

到此，主被叫资源预留完成（QCI1 EPS bearer 建立成功）：




```
v=0
o=- 108351810 1531652064 IN IP6 2409:8019:8430:3700:0000:0000:0000:FFF2
s=SBC call
c=IN IP6 2409:8019:8430:3700:0000:0000:0000:FFF2
t=0 0
m=audio 17092 RIP/AVP 107 101
b=AS:49
b=RR:1837
b=RS:612
a=rtpmap:107 AMR-WB/16000/1
a=fmtp:107 mode-change-capability=2;max-red=0
a=rtpmap:101 telephone-event/16000
a=fmtp:101 0-15
a=ptime:20
a=maxptime:240
a=sendrecv
a=curr:qos local sendrecv
a=curr:qos remote sendrecv
a=des:qos mandatory local sendrecv
a=des:qos mandatory remote sendrecv
```

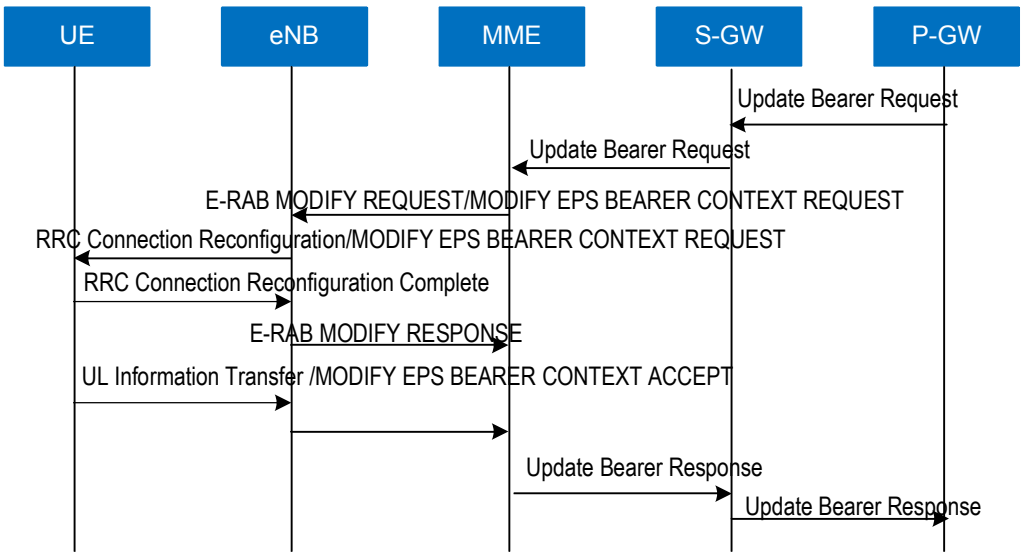
precondition

相关 IE 定义与 SIP INVITE 相同

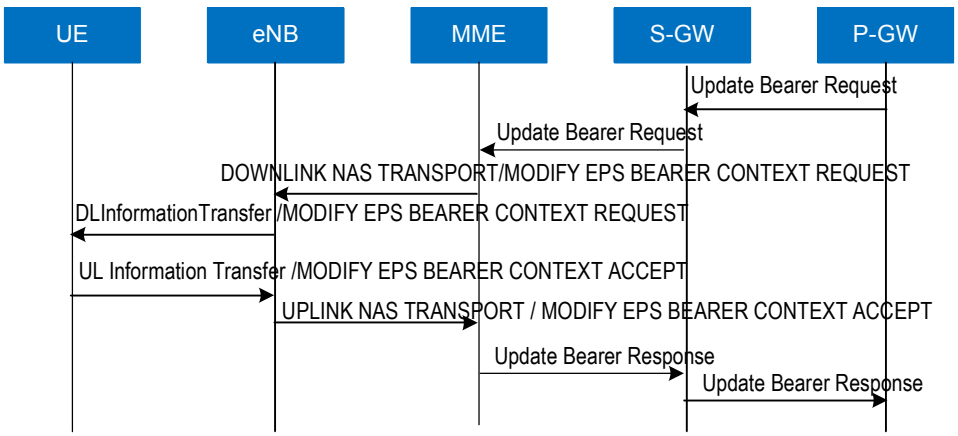
3.1.20 PDN GW initiated Dedicated bearer modification-Optional

PCRF 根据本地策略和业务媒体信息变更触发 P-GW 发起 QCI1 dedicated EPS bearer 修改，P-GW 根据修改内容不同，流程上略有差异：

（1） 有 QoS 参数更新（GBR/MBR/ARP）



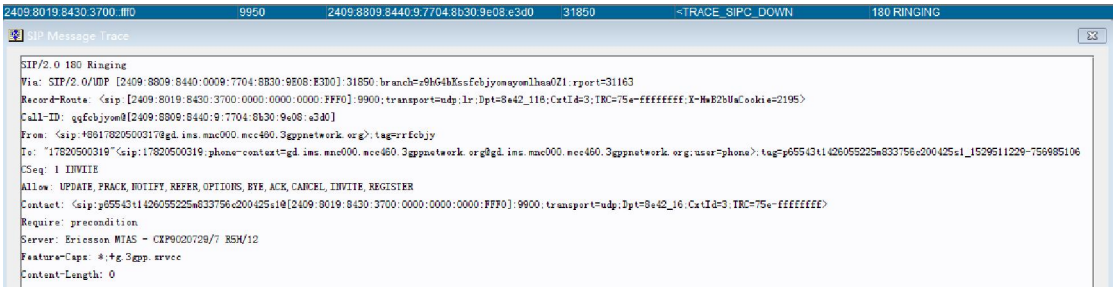
（2） 无 QoS 参数更新（TFT）



说明：本次呼叫流程中未涉及 dedicated bearer 更新流程。

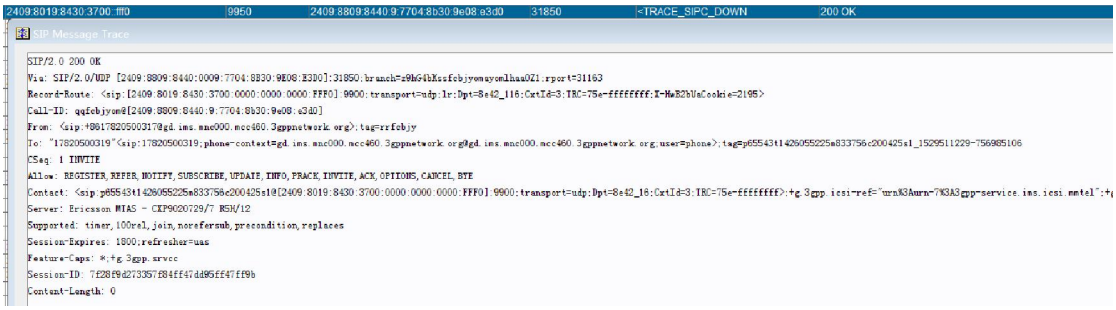
3.1.21 SIP 180 ringing

主被叫终端资源预留完成后，被叫侧振铃并发送 SIP 180 ringing。



3.1.22 SIP 200 OK（INVITE）

被叫摘机后，回复 SIP 200 OK（INVITE）给主叫：



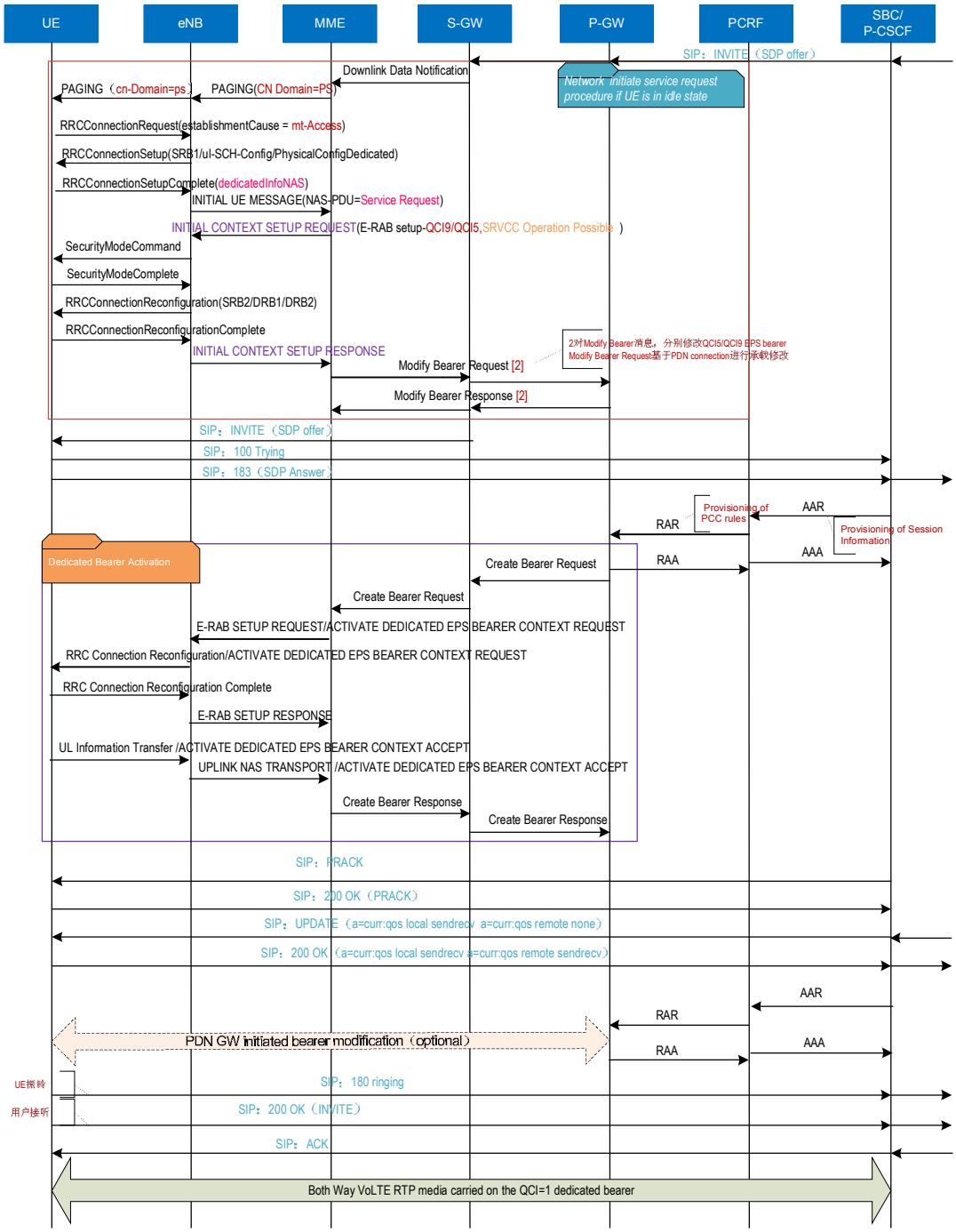
3.1.23 SIP ACK

主叫收到被叫摘机消息 SIP 200 OK（INVITE）后，回复 SIP ACK 消息给被叫进行确认。

此后，主被叫进入正常通话状态。



3.2 被叫呼叫建立流程

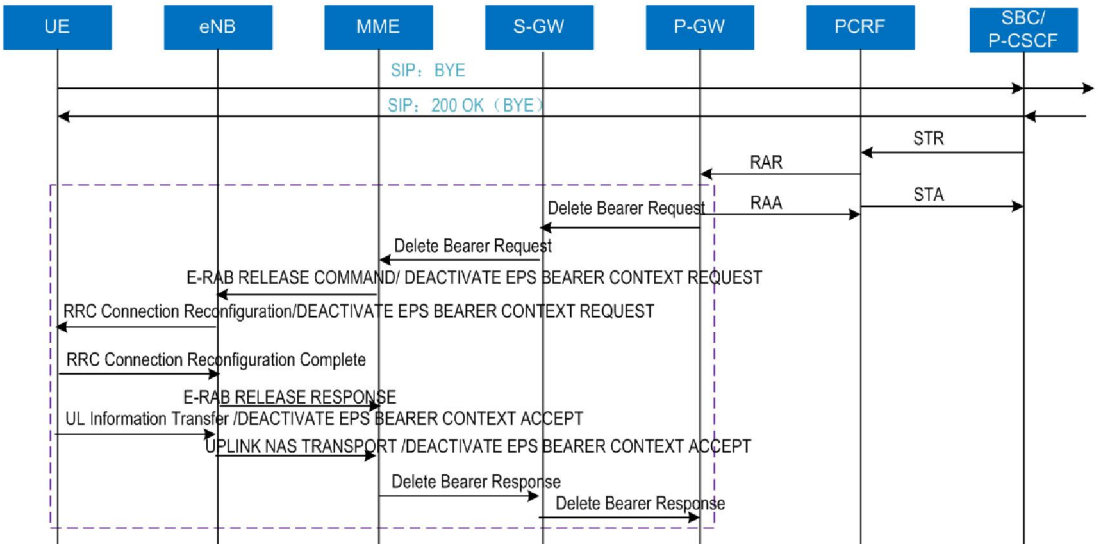


SBC 侧信令跟踪：

10.131.158.2	5060	10.132.170.2	16996	>TRACE_SIPC_UP	INVITE
10.132.170.2	16996	10.131.158.2	5060	<TRACE_SIPC_DOWN	100 TRYING
2409:8019:8430:3700::fff0	9900	2409:8809:8440:3:cdf8:f326:ac61:d29e	31846	<TRACE_SIPC_DOWN	INVITE
2409:8809:8440:3:cdf8:f326:ac61:d29e	31846	2409:8019:8430:3700::fff0	9950	>TRACE_SIPC_UP	100 TRYING
2409:8809:8440:3:cdf8:f326:ac61:d29e	31846	2409:8019:8430:3700::fff0	9950	>TRACE_SIPC_UP	183 SESSION PROGRESS
10.127.78.61	3869	10.127.71.46	7001	TRACE_DIAM_GQ	AAR
10.127.71.46	7001	10.127.78.61	3869	TRACE_DIAM_GQ	AAA
10.132.170.2	16996	10.131.158.2	5060	<TRACE_SIPC_DOWN	183 SESSION PROGRESS
10.131.158.2	5060	10.132.170.2	16996	>TRACE_SIPC_UP	PRACK
2409:8019:8430:3700::fff0	9900	2409:8809:8440:3:cdf8:f326:ac61:d29e	31846	<TRACE_SIPC_DOWN	PRACK
2409:8809:8440:3:cdf8:f326:ac61:d29e	31157	2409:8019:8430:3700::fff0	9900	>TRACE_SIPC_UP	200 OK
10.132.170.2	16996	10.131.158.2	5060	<TRACE_SIPC_DOWN	200 OK
10.131.158.2	5060	10.132.170.2	16996	>TRACE_SIPC_UP	UPDATE
2409:8019:8430:3700::fff0	9900	2409:8809:8440:3:cdf8:f326:ac61:d29e	31846	<TRACE_SIPC_DOWN	UPDATE
2409:8809:8440:3:cdf8:f326:ac61:d29e	31846	2409:8019:8430:3700::fff0	9950	>TRACE_SIPC_UP	200 OK
10.127.78.60	3868	10.127.71.42	7000	TRACE_DIAM_GQ	AAR
10.127.71.42	7000	10.127.78.60	3868	TRACE_DIAM_GQ	AAA
10.132.170.2	16996	10.131.158.2	5060	<TRACE_SIPC_DOWN	200 OK
2409:8809:8440:3:cdf8:f326:ac61:d29e	31846	2409:8019:8430:3700::fff0	9950	>TRACE_SIPC_UP	180 RINGING
10.132.170.2	16996	10.131.158.2	5060	<TRACE_SIPC_DOWN	180 RINGING
2409:8809:8440:3:cdf8:f326:ac61:d29e	31846	2409:8019:8430:3700::fff0	9950	>TRACE_SIPC_UP	200 OK
10.132.170.2	16996	10.131.158.2	5060	<TRACE_SIPC_DOWN	200 OK
10.131.158.2	5060	10.132.170.2	16996	>TRACE_SIPC_UP	ACK
2409:8019:8430:3700::fff0	9900	2409:8809:8440:3:cdf8:f326:ac61:d29e	31846	<TRACE_SIPC_DOWN	ACK

相关消息说明，请参考：主叫呼叫流程

3.3 VoLTE 呼叫释放



eNodeB 侧信令跟踪：

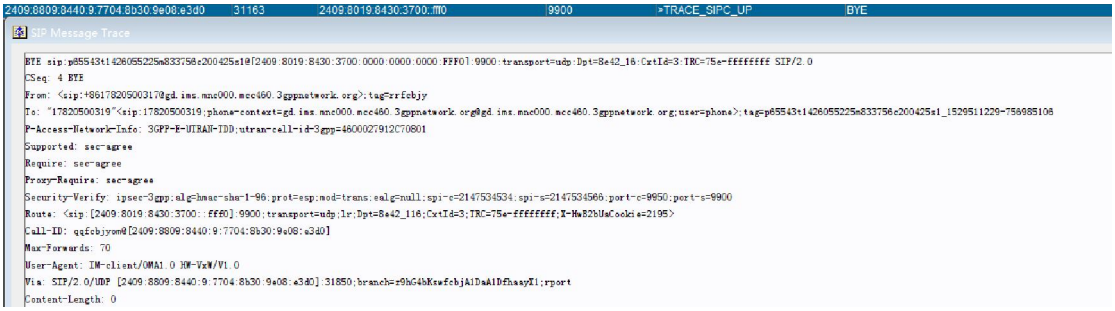
S1AP_ERAB_REL_CMD	Received From MME	cause=normal-release;
RRC_CONN_RECFG	Send to UE	
RRC_CONN_RECFG_CMP	Received From UE	
S1AP_ERAB_REL_RSP	Send to MME	
RRC_UL_INFO_TRANSF	Received From UE	
S1AP_UL_NAS_TRANS	Send to MME	

SBC 侧信令跟踪：

2409:8809:8440:9:7704:8b30:9e08:e3d0	31163	2409:8019:8430:3700::fff0	9900	>TRACE_SIPC_UP	BYE
10.132.170.2	16997	10.131.158.2	5060	<TRACE_SIPC_DOWN	BYE
10.131.158.2	5060	10.132.170.2	16996	>TRACE_SIPC_UP	BYE
2409:8019:8430:3700::fff0	9900	2409:8809:8440:3:cdf8:f326:ac61:d29e	31846	<TRACE_SIPC_DOWN	BYE
2409:8809:8440:3:cdf8:f326:ac61:d29e	31157	2409:8019:8430:3700::fff0	9900	>TRACE_SIPC_UP	200 OK
10.132.170.2	16996	10.131.158.2	5060	<TRACE_SIPC_DOWN	200 OK
10.131.158.2	5060	10.132.170.2	16997	>TRACE_SIPC_UP	200 OK
2409:8019:8430:3700::fff0	9950	2409:8809:8440:9:7704:8b30:9e08:e3d0	31850	<TRACE_SIPC_DOWN	200 OK
10.127.78.61	3869	10.127.71.46	7001	TRACE_DIAM_GQ	STR
10.127.71.46	7001	10.127.78.61	3869	TRACE_DIAM_GQ	STA

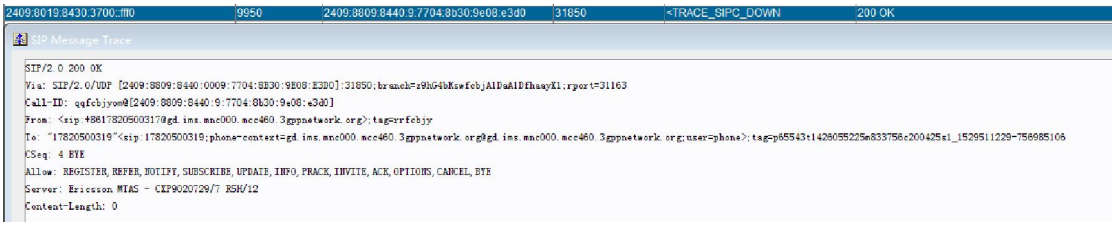
3.3.1 SIP BYE

用户挂机，发送 SIP BYE 消息：



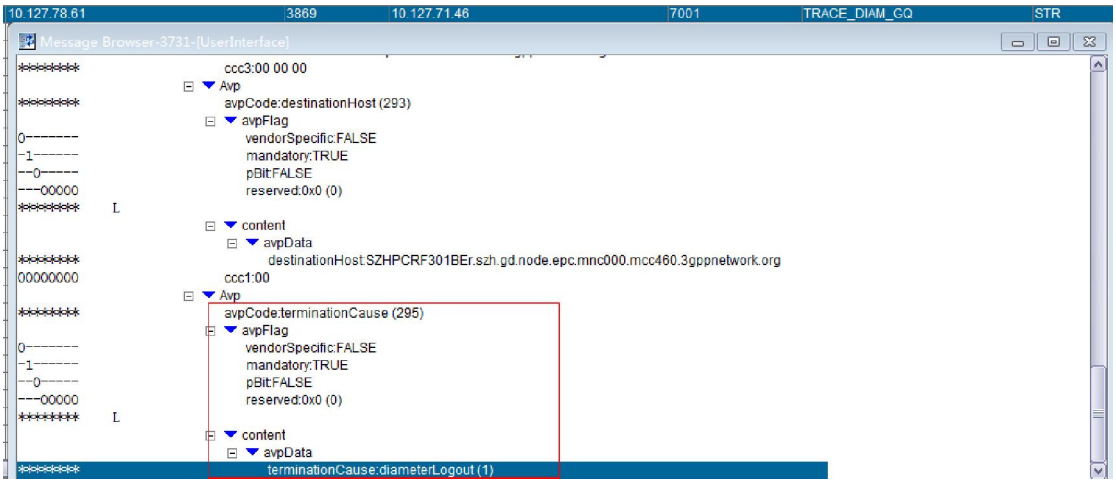
3.3.2 SIP 200 OK（BYE）

核心网转发 SIP BYE 消息，对端 UE 回复 SIP 200 OK，作为对 BYE 消息的响应，经过核心网发送给挂机用户所在 P-CSCF，进而转给挂机用户。

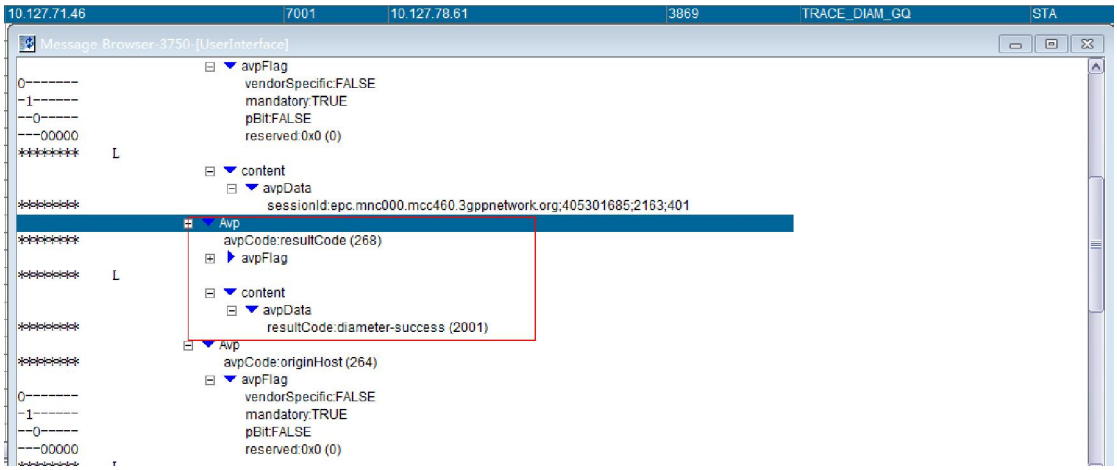


3.3.3 STR/STA

P-CSCF 发送 STR 消息触发 dedicated EPS bearer（QCI1）释放。



PCRF 回复 STA 消息对 STR 消息进行应答。



3.3.4 E-RAB RELEASE COMMAND

MME 发送 E-RAB RELEASE COMMAND 触发 E-RAB 释放（QCI1）

关键 IE	IE 取值定义
e-RABItem e-RAB-ID:0x7 (7) cause nas:normal-release (0)	36.413 9.1.3.5 E-RAB RELEASE COMMAND

IE/Group Name	Presence	Range	IE type and reference	Semantics description
Message Type	M		9.2.1.1	
MME UE S1AP ID	M		9.2.3.3	
eNB UE S1AP ID	M		9.2.3.4	
UE Aggregate Maximum Bit Rate	O		9.2.1.20	
E-RAB To Be Released List	M		E-RAB List 9.2.1.36	A value for E-RAB ID shall only be present once in E-RAB To Be Released List IE.

3.3.5 Deactivate EPS bearer context request

MME 在 E-RAB RELEASE COMMAND 消息中携带 NAS PDU（Deactivate EPS bearer context request），由 eNodeB 转发给 UE，触发释放相应的 EPS bearer。

关键 IE	IE 取值定义
no-security-protection-SM-message ePS-bearer-identity:0x7 (7) procedure-transaction-identity:0x0 (0) msg-body deactivateEPSBearerContextRequest eSM-cause:regular-deactivation (36)	24.301 8.3.12 Deactivate EPS bearer context request

IEI	Information Element	Type/Reference	Presence
	Protocol discriminator	Protocol discriminator 9.2	M
	EPS bearer identity	EPS bearer identity 9.3.2	M
	Procedure transaction identity	Procedure transaction identity 9.4	M
	Deactivate EPS bearer context request message identity	Message type 9.8	M
	ESM cause	ESM cause 9.9.4.4	M